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PHOSPHO-TAU ANTIBODIES AND METHODS OF USE

Abstract

Provided herein are compositions and methods relating to improved assays for establishing Alzheimer's disease. Further provided herein are compositions and methods comprising improved antibodies for assays including immunoassays.

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Background/Summary

CROSS-REFERENCE [0001] This application is a continuation of U.S. patent application Ser. No. 18/770,102 filed Jul. 11, 2024, which is a continuation of U.S. patent application Ser. No. 18/599,751 filed on Mar. 8, 2024, now U.S. Pat. No. 12,276,671, which is a continuation of International Application No.: PCT/US2022/042963 filed on Sep. 8, 2022, which claims the benefit of U.S. Provisional Patent Application No. 63/242,437 filed on Sep. 9, 2021, each of which is incorporated by reference in their entireties.

SEQUENCE LISTING

[0002] The instant application contains a sequence listing which has been submitted electronically in xml format and is hereby incorporated by reference in its entirety. Said xml copy, created on May 16, 2025, is named 58484-705_303_SL.xml and is 94,094 bytes in size.

BACKGROUND

[0003] The discovery of biomarkers and screening techniques of Alzheimer's disease (AD) and other tauopathies is an ongoing area of development in which these tools may be applied to screening populations to determine which non-demented individuals are at greatest risk of developing AD dementia and also to assess disease progression in patients. Proteins that are reflective of AD pathology, including amyloid beta 42 (A β .sub.42), neurofilament light chain, and various tau isoforms have been detected by a variety of means. Abnormal or excessive phosphorylation of tau has been associated with transformation of pathologically normal tau molecules into paired-helical-filament (PHF) tau and neurofibrillary tangles (NFTs) indicative of various tauopathy pathologies.

SUMMARY

[0004] Tau is an important microtubule-associated protein, abundantly expressed in CNS neurons, and serves critical roles in normal cellular physiology. Tau has also been found to be dysregulated in Alzheimer's disease and other tauopathies. Six isoforms of tau protein are generated from the TAU gene by alternative splicing. The isoforms differ from each other by the presence or absence of two N-terminal inserts and a repeat termed R2. All six protein isoforms of tau are highly soluble under normal and healthy cellular conditions and are typically regulated by phosphorylation and dephosphorylation. Tau has been demonstrated to interact with microtubules and promote microtubule assembly. In neurons, tau promotes the formation of axonal microtubules and stabilizes them. Tau has additional roles in driving neurite outgrowth. Impaired interaction of tau with microtubules may be an important component in the pathology, development, and progression of tauopathies. Hyperphosphorylation of tau is a hallmark feature of AD and other tauopathies and the extent of hyperphosphorylation is often correlated with disease progression.

Hyperphosphorylation of tau protein can result in the self-assembly of insoluble tangles of paired helical filaments and straight filaments of tau. These insoluble aggregates of tangles, termed neurofibrillary tangles (NFTs), are comprised of hyperphosphorylated tau and are considered to be pathological markers of tauopathies.

[0005] Phosphorylated tau (pTau), total tau, and A β .sub.42 each detected from the cerebrospinal fluid (CSF) and/or the blood are individual biomarkers for Alzheimer's disease and several other related tauopathies. CSF pTau is increased in individuals later confirmed to have AD both at the prodromal stages and the dementia stages compared to age- and gender-matched controls. CSF pTau levels exhibit a strong degree of correlation to the extent of cognitive impairment in individuals with AD. In fact, CSF pTau levels may be used with some degree of precision as a

biomarker to predict progression from cognitively unimpaired, to mild cognitive impairment (MCI) and then to AD dementia. In terms of utility as a biomarker to predict even relatively early stages of AD progression, CSF pTau has been shown to be significantly increased in samples from individuals with preclinical AD. Changes in the extent of pTau phosphorylation have been demonstrated in both preclinical sporadic cases of AD and in early stages of autosomal-dominant AD. Blood levels of pTau, total tau, and A β .sub.42 are generally lower than CSF levels when assayed within the same individual and may be utilized as informative biomarkers for AD and other related tauopathies if blood levels of these biomarkers can be assayed with sufficient specificity and precision.

[0006] Several sites of phosphorylation contributing to hyperphosphorylated tau which aggregates into NFTs have been identified. In the longest tau isoform, 79 potential serine or threonine phosphorylation sites are present and at least 30 of these sites have been identified as phosphorylated in NFT aggregates. A common site used to assay tau molecules for phosphorylation status is at threonine-181. CSF fluid contains an array of tau fragments at various abundances. Fragments of tau from the N-terminal region and from the middle region of tau polypeptides are considerably more abundant in CSF samples than C-terminal tau fragments. Plasma samples from individuals also contain tau polypeptides and tau polypeptide fragments, however they tend to be present at lower concentrations than in matched CSF samples. Being able to detect tau phosphorylation at particular amino acid residues relevant for disease pathology and progression is a critical component of diagnosis, disease staging, and as a metric to measure treatment efficacy for AD and other tauopathies. Detection and measurement of pTau levels at particular disease-relevant residues from plasma samples would aid greatly to the development of more sensitive and finely-tuned diagnosis, prognosis, and disease analysis for individuals who may be at risk for developing or are at early stages of AD or other tauopathies. Phosphorylation of tau at threonine 217 (pTau 217) is one such residue of particular interest in development new biomarkers and diagnostic assays. Alterations in pTau biomarker concentration in CSF and in plasma are thought to precede measurable behavioral or cognitive changes in AD and in other tauopathies. A development of new assays to enable a continuum of specific points and extents of tau phosphorylation of certain residues would undoubtedly aid in the clinically relevant medical diagnosis and treatment decisions. A comparison of results from new assays to results from existing assays can also yield further medically informative determinations. Results from plasma-based tau biomarker assays can be compared against matched CSF samples (detecting CSF pTau or CSF soluble $\Delta\beta$) and also against positron emission tomography (PET) scans detecting an extent and locations of $\Delta\beta$ aggregates as metrics for their utility, especially for analysis at preclinical or early disease stages.

[0007] Provided herein are methods for detecting phosphorylated tau in a sample from an individual comprising: performing an immunoassay on the sample using an antibody or antibody fragment comprising a variable domain, heavy chain region (VH) and a variable domain, light chain region (VL), wherein the VH comprises an amino acid sequence at least about 90% identical to a sequence as set forth in any one of SEQ ID NOs: 30-34, and wherein the VL comprises an amino acid sequence at least about 90% identical to a sequence as set forth in any one of SEQ ID NOs: 35-40. Further provided herein are methods for detecting phosphorylated tau in a sample from an individual, wherein the phosphorylated tau is selected from the group consisting of pTau-181, pTau-212, pTau-217, pTau-231, pTau-214, and pTau-220. Further provided herein are methods for detecting phosphorylated tau in a sample from an individual, wherein the phosphorylated tau is pTau-217. Further provided herein are methods for detecting phosphorylated tau in a sample from an individual, wherein the phosphorylated tau is pTau-231. Further provided herein are methods for detecting phosphorylated tau in a sample from an individual, wherein the method detects pTau-217 and pTau-231. Further provided herein are methods for detecting phosphorylated tau in a sample from an individual, wherein the method detects pTau-212 and pTau-217. Further provided herein are methods for detecting phosphorylated tau in a sample from an individual, wherein the method

[illegible]

and wherein the VL comprises an amino acid sequence at least about 90% identical to SEQ ID NO: 40. Further provided herein are methods for detecting phosphorylated tau in a sample from an individual, wherein the antibody or antibody fragment comprises an amino acid sequence at least about 90% identical to any one of SEQ ID NOs: 41-51. Further provided herein are methods for detecting phosphorylated tau in a sample from an individual, further comprising performing an assay on the sample to determine a level of a biomarker selected from the group consisting of A β 42, A β 40, A β 38, BACE1, hFABP, TREM2, YKL-40, IP-10, neurogranin, SNAP-25, synaptotagmin, alpha-synuclein, TDP-43, ferritin, VILIP-1, NfL, GFAP, and combinations thereof. Further provided herein are methods for detecting phosphorylated tau in a sample from an individual, wherein the sample is selected from the group consisting of a blood sample, a plasma sample, a serum sample, and a cerebrospinal fluid (CSF) sample. Further provided herein are methods for detecting phosphorylated tau in a sample from an individual, further comprising establishing Alzheimer's disease in the individual based on detection of phosphorylated tau. Further provided herein are methods for detecting phosphorylated tau in a sample from an individual, further comprising establishing prognosis of the individual for developing Alzheimer's disease based on detection of phosphorylated tau. Further provided herein are methods for detecting phosphorylated tau in a sample from an individual, further determining the individual's age, genotype, or expression of a biomarker. Further provided herein are methods for detecting phosphorylated tau in a sample from an individual, wherein the biomarker is selected from the group consisting of A042, A040, A038, BACE1, hFABP, TREM2, YKL-40, IP-10, neurogranin, SNAP-25, synaptotagmin, alpha-synuclein, TDP-43, ferritin, VILIP-1, NfL, GFAP, and combinations thereof. Further provided herein are methods for detecting phosphorylated tau in a sample from an individual, wherein the method has a specificity of at least about 80% for detecting phosphorylated tau. Further provided herein are methods for detecting phosphorylated tau in a sample from an individual, wherein the method has a specificity of at least about 85% for detecting phosphorylated tau. Further provided herein are methods for detecting phosphorylated tau in a sample from an individual, wherein the method has a specificity of at least about 90% for detecting phosphorylated tau. Further provided herein are methods for detecting phosphorylated tau in a sample from an individual, wherein the method has a sensitivity of at least about 80% for detecting phosphorylated tau. Further provided herein are methods for detecting phosphorylated tau in a sample from an individual, wherein the method has a sensitivity of at least about 85% for detecting phosphorylated tau. Further provided herein are methods for detecting phosphorylated tau in a sample from an individual, wherein the method has a sensitivity of at least about 90% for detecting phosphorylated tau. Further provided herein are methods for detecting phosphorylated tau in a sample from an individual, wherein the method is capable of detecting phosphorylated tau in the sample at a limit of detection of at least about 1.0 picogram per milliliter (pg/mL). Further provided herein are methods for detecting phosphorylated tau in a sample from an individual, wherein the method is capable of detecting phosphorylated tau in the sample at a limit of detection of at least about 1.5 picogram per milliliter (pg/mL). Further provided herein are methods for detecting phosphorylated tau in a sample from an individual, wherein the method is capable of detecting phosphorylated tau in the sample at a limit of detection of at least about 5 picogram per milliliter (pg/mL).

[0008] Also provided herein are, in certain embodiments, anti-tau antibodies comprising i) a heavy chain comprising variable heavy chain (VH) domain and ii) a light chain comprising a variable light chain (VL) domain, wherein the VH domain comprises HCDR1 sequence comprising a sequence selected from SEQ ID NOs: 1-5, HCDR2 sequence comprising a sequence selected from SEQ ID NOs: 6-9, and HCDR3 sequence comprising a sequence selected from SEQ ID NOs: 10-13, and VL domain comprises LCDR1 sequence comprising a sequence selected from SEQ ID NOs: 14-19, LCDR2 sequence comprising a sequence selected from SEQ ID NOs: 20-23, and LCDR3 sequence comprising a sequence selected from SEQ ID NOs: 24-29. In some

embodiments, the HCDR1 sequence comprises SEQ ID NO: 1, the HCDR2 sequence comprises SEQ ID NO: 6, the HCDR3 sequence comprises SEQ ID NO: 10, the LCDR1 sequence comprises SEQ ID NO: 14, the LCDR2 sequence comprises SEQ ID NO: 20, and the LCDR3 sequence comprises SEQ ID NO: 24. In some embodiments, the HCDR1 sequence comprises SEQ ID NO: 2, the HCDR2 sequence comprises SEQ ID NO: 7, the HCDR3 sequence comprises SEQ ID NO: 11, the LCDR1 sequence comprises SEQ ID NO: 15, the LCDR2 sequence comprises SEQ ID NO: 21, and the LCDR3 sequence comprises SEQ ID NO: 25. In some embodiments, the HCDR1 sequence comprises SEQ ID NO: 2, the HCDR2 sequence comprises SEQ ID NO: 7, the HCDR3 sequence comprises SEQ ID NO: 11, the LCDR1 sequence comprises SEQ ID NO: 16, the LCDR2 sequence comprises SEQ ID NO: 22, and the LCDR3 sequence comprises SEQ ID NO: 26. In some embodiments, the HCDR1 sequence comprises SEQ ID NO: 3, the HCDR2 sequence comprises SEQ ID NO: 8, the HCDR3 sequence comprises SEQ ID NO: 10, the LCDR1 sequence comprises SEQ ID NO: 17, the LCDR2 sequence comprises SEQ ID NO: 20, and the LCDR3 sequence comprises SEQ ID NO: 27. In some embodiments, the HCDR1 sequence comprises SEQ ID NO: 4, the HCDR2 sequence comprises SEQ ID NO: 7, the HCDR3 sequence comprises SEQ ID NO: 12, the LCDR1 sequence comprises SEQ ID NO: 18, the LCDR2 sequence comprises SEQ ID NO: 23, and the LCDR3 sequence comprises SEQ ID NO: 28. In some embodiments, the HCDR1 sequence comprises SEQ ID NO: 5, the HCDR2 sequence comprises SEQ ID NO: 9, the HCDR3 sequence comprises SEQ ID NO: 13, the LCDR1 sequence comprises SEQ ID NO: 19, the LCDR2 sequence comprises SEQ ID NO: 21, and the LCDR3 sequence comprises SEQ ID NO: 29. Further provided herein are, in some embodiments, anti-tau antibodies comprising i) a heavy chain comprising variable heavy chain (VH) domain and ii) a light chain comprising a variable light chain (VL) domain, wherein the VH domain comprises at least 80%, at least 85%, at least 90%, at least 95% sequence identity to a sequence selected from SEQ ID NOs: 30-34. Further provided herein are, in some embodiments, anti-tau antibodies comprising i) a heavy chain comprising variable heavy chain (VH) domain and ii) a light chain comprising a variable light chain (VL) domain, wherein the VL domain comprises at least 80%, at least 85%, at least 90%, at least 95% sequence identity to a sequence selected from SEQ ID NOs: 35-40. In some embodiments, the anti-tau antibody described herein is a chimeric antibody or antigen binding fragment thereof. In some embodiments, the anti-tau antibody described herein comprises an IgG-scFv, nanobody, BiTE, diabody, DART, TandAb, scDiabody, scDiabody-CH3, triple body, mini-antibody, minibody, TriBi minibody, scFv-CH3 KIH, Fab-scFv-Fc KIH, Fab-scFv, scFv-CH-CL-scFv, Fab', F(ab')₂, F(ab')₃, F(ab')₂-scFv₂, scFv, scFv-KIH, Fab-scFv-Fc, tetravalent HCAb, scDiabody-Fc, diabody-Fc, tandem scFv-Fc, or intrabody. In some embodiments, the anti-tau antibody described herein is an IgG1 antibody. In some embodiments, the anti-tau antibody described herein is an IgG2 antibody. In some embodiments, the anti-tau antibody described herein is an IgG4 antibody. Further provided herein are, in some embodiments, anti-tau antibodies comprising i) a heavy chain comprising variable heavy chain (VH) domain and ii) a light chain comprising a variable light chain (VL) domain, wherein the light chain is a kappa chain. Further provided herein are, in some embodiments, anti-tau antibodies comprising i) a heavy chain comprising variable heavy chain (VH) domain and ii) a light chain comprising a variable light chain (VL) domain, wherein the anti-tau antibody has a binding affinity to human tau of about 100 pM to about 3 nM. Provided herein are, in some embodiments, anti-tau antibodies comprising a VH domain that is encoded by a nucleic acid comprising at least 80%, at least 85%, at least 90%, at least 95% sequence identity to a sequence selected from SEQ ID NOs: 52-56. Provided herein are, in some embodiments, anti-tau antibodies comprising a VL domain that is encoded by a nucleic acid comprising at least 80%, at least 85%, at least 90%, at least 95% sequence identity to a sequence selected from SEQ ID NOs: 57-62. Provided herein are, in some embodiments, anti-tau antibodies comprising a VH domain that is encoded by a nucleic acid comprising at least 80%, at least 85%, at least 90%, at least 95% sequence identity to a sequence selected from SEQ ID NOs:

52-56 and a VL domain that is encoded by a nucleic acid comprising at least 80%, at least 85%, at least 90%, at least 95% sequence identity to a sequence selected from SEQ ID NOs: 57-62. Provided herein are, in some embodiments, anti-tau antibodies comprising a VH domain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NOs: 52-56. Provided herein are, in some embodiments, anti-tau antibodies comprising a VL domain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NOs: 57-62. Provided herein are, in some embodiments, anti-tau antibodies comprising a VH domain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NOs: 52-56 and a VL domain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NOs: 57-62.

INCORPORATION BY REFERENCE

[0009] All publications, patents, and patent applications mentioned in this specification are herein incorporated by reference to the same extent as if each individual publication, patent, or patent application was specifically and individually indicated to be incorporated by reference.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1 depicts a schema of the single molecule array (Simoa®) method used herein for assaying tau antibodies described herein. After substrate is added to sample (sandwich ELISA on bead, 1.1), sample is added to Simoa® Disk (1.2). Beads are given time to settle into microarray wells on disk (one bead per well) (1.3). Then, sealing oil is used to remove excess beads to allow for imaging (1.4). Beads that have sandwich complex (positive beads) will fluoresce with the substrate and show up during imaging; beads without sandwich complex (negative) will still show up in imaging but will not fluoresce (1.5). The percentage of positive beads is converted to an AEB (average enzymes per bead) value.

[0011] FIGS. 2A-2D depict data for Antibody 1, Antibody 2, Antibody 3, Antibody 4, Antibody 5, and Antibody 6 in the Simoa® assay.

[0012] FIG. 3 depict ELISA data.

[0013] FIGS. 4A-4G depict data for immunohistochemistry staining of Antibody 6.

[0014] FIGS. 5A-5G depict data for immunohistochemistry staining of Antibody 5.

[0015] FIGS. 6A-6G depict data for immunohistochemistry staining of Antibody 2.

[0016] FIG. 7 depicts a diagram of an indirect ELISA assay and graphs of ELISA data assaying antibody binding to pTau-217 peptides.

[0017] FIG. 8 depicts a graph of signal/noise (S/N) analysis of ELISA assay for Antibody 2 binding to a pTau-217 peptide for 120 clinic samples derived from plasma and a graph of a coefficient of variation (CV %) for ELISA assay for Antibody 2 binding to a pTau-217 peptide for 120 clinic samples derived from plasma.

[0018] FIG. 9 depicts graphs of calibration curves (Cal curves) for a Simoa®-based pTau-217 assay using Antibody 2 on groups (plates) designated QTx of clinical samples derived from cerebrospinal fluid (68 CSF samples) and plasma (120 plasma samples) compared to the assay using ADx p204 antibody.

[0019] FIG. 10 depicts a graph of Simoa®-based pTau assay-217 results using Antibody 2 in matched samples from the sample individual derived from either plasma (Y-axis) or CSF (X-axis) and statistical analysis of correlated results in individual with a clinical diagnosis of either non-Alzheimer's disease, an uncertain diagnosis, or Alzheimer's disease.

[0020] FIG. 11 depicts a graph of Simoa®-based pTau assay-217 results using Antibody 2 per sample vs Simoa®-based pTau assay-181 results using Antibody 2 per sample and statistical analysis of correlated results.

[0021] FIG. 12 depicts a graph of Simoa®-based pTau assay-217 results using Antibody 2 per

sample vs Simoa®-based Tau assay results using Innotech pTau 181 antibody per sample and statistical analysis of correlated results.

[0022] FIG. **13** depicts a graph of Simoa®-based pTau assay results using Antibody 2 as a capture antibody, antibody ADx p204 as a detector antibody and a peptide as calibrator and statistical analysis of correlated results.

[0023] FIG. **14** depicts a graph of Simoa®-based pTau assay-217 results using Antibody 2 grouping together samples from individuals with a clinical diagnosis of Alzheimer's disease and samples from control individuals derived from either CSF or plasma.

[0024] FIG. **15** depicts a graph of Simoa®-based pTau assay-217 results using Antibody 2 on various concentrations of EDTA plasma samples and a chart of listing coefficient of variation for each sample concentration to illustrate the precision of the assay.

[0025] FIG. **16** depicts graphs of Simoa®-based pTau assay-217 results using Antibody 2 graphed as coefficient of variation (CV %) vs measured concentration.

[0026] FIG. **17** depicts a graph of Simoa®-based pTau assay-217 results using Antibody 2 and statistical analysis of parallelism with determines whether actual samples containing high endogenous analyte concentrations provide a similar degree of detection in a standard curve after dilutions.

[0027] FIG. **18** depicts a graph of Simoa®-based pTau assay-217 results using Antibody 2 and statistical analysis of linearity with determines whether sample matrices spiked with detection analyte above an upper limit of detect can still provide reliable quantification after dilution within standard curve ranges for four samples plus a buffer spike.

[0028] FIG. **19** depicts a graph of Simoa®-based pTau assay-217 results using Antibody 2 and statistical analysis of linearity with determines whether sample matrices spiked with detection analyte above an upper limit of detect can still provide reliable quantification after dilution within standard curve ranges for three samples plus a calibration sample.

[0029] FIG. **20** depicts a graph of Simoa®-based pTau assay-217 results using Antibody 2 in a clinical validation of a memory clinic cohort and a graph of receiver-operating characteristic (ROC) analysis graphed against pTau-217 assay sensitivity.

[0030] FIG. **21** depicts a graph of Simoa®-based pTau assay-217 results using Antibody 2 on groups from Control and AD dementia individuals.

[0031] FIG. **22** depicts a graph of Simoa®-based pTau assay-181 results using an antibody from Quanterix® on groups from Control and AD dementia individuals and a chart of sample stratification.

[0032] FIG. **23** depicts graphs of Simoa®-based pTau assay-217 results using Antibody 2, and Simoa®-based pTau assay-181 results using an antibody from Quanterix® showing precision plots with calculated coefficient of variation.

[0033] FIG. **24** depicts a graph of clinical performance of various pTau Simoa®-based assays comparing sensitivity and specificity and a chart with a statistical analysis of results.

[0034] FIG. **25** depicts a schematic diagram of Tau indicating the relative location of various protein domains and the locations of threonine residues which can be assayed for phosphorylation status using methods disclosed herein.

[0035] FIG. **26** depicts graphs of reactivity to a Tau fragment with non-phosphorylated T217 (Bio-pt654) and full length Tau (Tau441) in indirect ELISA for various antibodies.

[0036] FIG. **27** depicts graphs of reactivity to Tau fragments with phosphorylated T181 (Bio-pt126) and phosphorylated T231 (Bio-pt146) in indirect ELISA for various antibodies.

[0037] FIG. **28** depicts a diagram of an assay utilizing a pTau217 monoclonal antibody as a capture tool for various synthetic peptides and a graph of results for this assay using Antibody 2 as the capture tool.

[0038] FIG. **29** depicts Western blot analysis using various Tau antibodies on brain lysate samples from AD patients or control subjects.

DETAILED DESCRIPTION

[0039] Alzheimer's disease (AD) is a complex disease and effective treatment requires accurate diagnosis. Described herein are improved compositions and methods for detecting AD that comprises improved antibodies for use in diagnostic and/or prognostic assays.

Certain Terminologies

[0040] Throughout this disclosure, various embodiments are presented in a range format. It should be understood that the description in range format is merely for convenience and brevity and should not be construed as an inflexible limitation on the scope of any embodiments. Accordingly, the description of a range should be considered to have specifically disclosed all the possible subranges as well as individual numerical values within that range to the tenth of the unit of the lower limit unless the context clearly dictates otherwise. For example, description of a range such as from 1 to 6 should be considered to have specifically disclosed subranges such as from 1 to 3, from 1 to 4, from 1 to 5, from 2 to 4, from 2 to 6, from 3 to 6 etc., as well as individual values within that range, for example, 1.1, 2, 2.3, 5, and 5.9. This applies regardless of the breadth of the range. The upper and lower limits of these intervening ranges may independently be included in the smaller ranges, and are also encompassed within the invention, subject to any specifically excluded limit in the stated range. Where the stated range includes one or both of the limits, ranges excluding either or both of those included limits are also included in the invention, unless the context clearly dictates otherwise.

[0041] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of any embodiment. As used herein, the singular forms “a,” “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

[0042] Unless specifically stated or obvious from context, as used herein, the term “about” in reference to a number or range of numbers is understood to mean the stated number and numbers $\pm 10\%$ thereof, or 10% below the lower listed limit and 10% above the higher listed limit for the values listed for a range.

[0043] The terms “individual,” “patient,” or “subject” are used interchangeably. None of the terms require or are limited to a situation characterized by the supervision (e.g., constant or intermittent) of a health care worker (e.g., a doctor, a registered nurse, a nurse practitioner, a physician's assistant, an orderly, or a hospice worker). Further, these terms refer to human or animal subjects.

[0044] The term “antibody” herein is used in the broadest sense and includes monoclonal antibodies, including intact antibodies and functional (antigen-binding) antibody fragments thereof, including fragment antigen binding (Fab) fragments, F(ab')₂ fragments, Fab' fragments, Fv fragments, recombinant IgG (rIgG) fragments, single chain antibody fragments, including single chain variable fragments (sFv or scFv), and single domain antibodies (e.g., sdAb, sdFv, nanobody) fragments. The term encompasses genetically engineered and/or otherwise modified forms of immunoglobulins, such as intrabodies, peptibodies, chimeric antibodies, and heteroconjugate antibodies, tandem di-scFv, tandem tri-scFv. Unless otherwise stated, the term “antibody” should be understood to encompass functional antibody fragments thereof. The term also encompasses intact or full-length antibodies, including antibodies of any class or sub-class, including IgG and sub-classes thereof, IgM, IgE, IgA, and IgD. The antibody can comprise a rabbit IgG1 constant region. The antibody can comprise a rabbit IgG4 constant region. An antibody includes, but is not limited to, full-length and native antibodies, as well as fragments and portion thereof retaining the binding specificities thereof, such as any specific binding portion thereof including those having any number of, immunoglobulin classes and/or isotypes (e.g., IgG1, IgG2, IgG3, IgG4, IgM, IgA,

IgD, IgE and IgM); and biologically relevant (antigen-binding) fragments or specific binding portions thereof, including but not limited to Fab, F(ab')₂, Fv, and scFv (single chain or related entity). A monoclonal antibody is generally one within a composition of substantially homogeneous antibodies; thus, any individual antibodies comprised within the monoclonal antibody composition are identical except for possible naturally occurring mutations that may be present in minor amounts. A monoclonal antibody can comprise a rabbit IgG1 constant region or a rabbit IgG4 constant region.

[0045] The term “complementarity determining region” or “CDR” is a segment of the variable region of an antibody that is complementary in structure to the epitope to which the antibody binds and is more variable than the rest of the variable region. Accordingly, a CDR is sometimes referred to as hypervariable region. A variable region comprises three CDRs. CDR peptides can be obtained by constructing genes encoding the CDR of an antibody of interest. Such genes are prepared, for example, by using the polymerase chain reaction to synthesize the variable region from RNA of antibody-producing cells. See, for example, Larrick et al., *Methods: A Companion to Methods in Enzymology* 2: 106 (1991); Courtenay-Luck, “Genetic Manipulation of Monoclonal Antibodies,” in *Monoclonal Antibodies: Production, Engineering and Clinical Application*, Ritter et al. (eds.), pages 166-179 (Cambridge University Press 1995); and Ward et al., “Genetic Manipulation and Expression of Antibodies,” in *Monoclonal Antibodies: Principles and Applications*, Birch et al., (eds.), pages 137-185 (Wiley-Liss, Inc. 1995).

[0046] The term “Fab” refers to a protein that contains the constant domain of the light chain and the first constant domain (CH1) of the heavy chain. Fab fragments differ from Fab' fragments by the addition of a few residues at the carboxy terminus of the heavy chain CH1 domain including one or more cysteines from the antibody hinge region. Fab'-SH is the designation herein for Fab' in which the cysteine residue(s) of the constant domains bear a free thiol group. Fab' fragments are produced by reducing the F(ab')₂ fragment's heavy chain disulfide bridge. Other chemical couplings of antibody fragments are also known.

[0047] A “single-chain variable fragment (scFv)” is a fusion protein of the variable regions of the heavy (VH) and light chains (VL) of an antibody, connected with a short linker peptide of ten to about 25 amino acids. The linker is usually rich in glycine for flexibility, as well as serine or threonine for solubility, and can either connect the N-terminus of the VH with the C-terminus of the VL, or vice versa. This protein retains the specificity of the original antibody, despite removal of the constant regions and the introduction of the linker. scFv antibodies are, e.g. described in Houston, J. S., *Methods in Enzymol.* 203 (1991) 46-96). In addition, antibody fragments comprise single chain polypeptides having the characteristics of a VH domain, namely being able to assemble together with a VL domain, or of a VL domain, namely being able to assemble together with a VH domain to a functional antigen binding site and thereby providing the antigen binding property of full length antibodies.

[0048] As used herein, the term “percent (%) amino acid sequence identity” with respect to a sequence is defined as the percentage of amino acid residues in a candidate sequence that are identical with the amino acid residues in the specific sequence, after aligning the sequences and introducing gaps, if necessary, to achieve the maximum percent sequence identity, and not considering any conservative substitutions as part of the sequence identity. Alignment for purposes of determining percent amino acid sequence identity can be achieved in various ways that are within the skill in the art, for instance, using publicly available computer software such as EMBOSS MATCHER, EMBOSS WATER, EMBOSS STRETCHER, EMBOSS NEEDLE, EMBOSS LALIGN, BLAST, BLAST-2, ALIGN or Megalign (DNASTAR) software. Those skilled in the art can determine appropriate parameters for measuring alignment, including any algorithms needed to achieve maximal alignment over the full length of the sequences being compared.

[0049] In situations where ALIGN-2 is employed for amino acid sequence comparisons, the % amino acid sequence identity of a given amino acid sequence A to, with, or against a given amino

acid sequence B (which can alternatively be phrased as a given amino acid sequence A that has or comprises a certain % amino acid sequence identity to, with, or against a given amino acid sequence B) is calculated as follows: 100 times the fraction X/Y, where X is the number of amino acid residues scored as identical matches by the sequence alignment program ALIGN-2 in that program's alignment of A and B, and where Y is the total number of amino acid residues in B. It will be appreciated that where the length of amino acid sequence A is not equal to the length of amino acid sequence B, the % amino acid sequence identity of A to B will not equal the % amino acid sequence identity of B to A. Unless specifically stated otherwise, all % amino acid sequence identity values used herein are obtained as described in the immediately preceding paragraph using the ALIGN-2 computer program.

[0050] The terms “complementarity determining region,” and “CDR,” which are synonymous with “hypervariable region” or “HVR,” are known in the art to refer to non-contiguous sequences of amino acids within antibody variable regions, which confer antigen specificity and/or binding affinity. In general, there are three CDRs in each heavy chain variable region (CDR-H1, CDR-H2, CDR-H3) and three CDRs in each light chain variable region (CDR-L1, CDR-L2, CDR-L3). “Framework regions” and “FR” are known in the art to refer to the non-CDR portions of the variable regions of the heavy and light chains. In general, there are four FRs in each full-length heavy chain variable region (FR-H1, FR-H2, FR-H3, and FR-H4), and four FRs in each full-length light chain variable region (FR-L1, FR-L2, FR-L3, and FR-L4). The precise amino acid sequence boundaries of a given CDR or FR can be readily determined using any of a number of well-known schemes, including those described by Kabat et al. (1991), “Sequences of Proteins of Immunological Interest,” 5th Ed. Public Health Service, National Institutes of Health, Bethesda, MD (“Kabat” numbering scheme), Al-Lazikani et al., (1997) JMB 273,927-948 (“Chothia” numbering scheme); MacCallum et al., J. Mol. Biol. 262:732-745 (1996), “Antibody-antigen interactions: Contact analysis and binding site topography,” J. Mol. Biol. 262, 732-745.” (“Contact” numbering scheme); Lefranc M P et al., “IMGT unique numbering for immunoglobulin and T cell receptor variable domains and Ig superfamily V-like domains,” Dev Comp Immunol, 2003 January; 27(1):55-77 (“IMGT” numbering scheme); Honegger A and Plückthun A, “Yet another numbering scheme for immunoglobulin variable domains: an automatic modeling and analysis tool,” J Mol Biol, 2001 Jun. 8; 309(3):657-70, (“Aho” numbering scheme); and Whitelegg N R and Rees A R, “WAM: an improved algorithm for modelling antibodies on the WEB,” Protein Eng. 2000 December; 13(12):819-24 (“AbM” numbering scheme. In certain embodiments the CDRs of the antibodies described herein can be defined by a method selected from Kabat, Chothia, IMGT, Aho, AbM, or combinations thereof.

[0051] The boundaries of a given CDR or FR may vary depending on the scheme used for identification. For example, the Kabat scheme is based on structural alignments, while the Chothia scheme is based on structural information. Numbering for both the Kabat and Chothia schemes is based upon the most common antibody region sequence lengths, with insertions accommodated by insertion letters, for example, “30a,” and deletions appearing in some antibodies. The two schemes place certain insertions and deletions (“indels”) at different positions, resulting in differential numbering. The Contact scheme is based on analysis of complex crystal structures and is similar in many respects to the Chothia numbering scheme.

[0052] Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which the methods and compositions described herein belong. Although any methods and materials similar or equivalent to those described herein can also be used in the practice or testing of the methods and compositions described herein, representative illustrative methods and materials are now described.

Tau Antibodies

[0053] Provided herein are antibodies that bind to tau. In some instances, the antibodies that bind to tau are monoclonal antibodies. In certain aspects, disclosed herein is an anti-tau antibody. In some

instances, the anti-tau antibody specifically binds to mammalian tau. In some instances, the anti-tau antibody specifically binds to a human tau. In some instances, the anti-tau antibody specifically binds to an N-terminal portion of tau. In some instances, the anti-tau antibody specifically binds to an N-terminal portion of human tau. In some instances, the anti-tau antibody specifically binds to an portion of tau comprising protein domain P2. In some instances, the anti-tau antibody specifically binds to an portion of human tau comprising protein domain P2. In some instances, the anti-tau antibody specifically binds to an portion of tau comprising protein domain P1. In some instances, the anti-tau antibody specifically binds to an portion of human tau comprising protein domain P1. In some instances, the anti-tau antibody specifically binds to an portion of tau comprising protein domains P1 and P2. In some instances, the anti-tau antibody specifically binds to an portion of human tau comprising protein domains P1 and P2.

[0054] In some embodiments, the anti-tau antibody comprises i) a heavy chain comprising a variable heavy chain (VH) domain and ii) a light chain comprising a variable light chain (VL) domain. In some embodiments, VH domain comprises heavy chain CDR1 (HCDR1) sequence comprising a sequence selected from SEQ ID NOs: 1-5, heavy chain CDR2 (HCDR2) sequence comprising a sequence selected from SEQ ID NOs: 6-9, and heavy chain CDR3 (HCDR3) sequence comprising a sequence selected from SEQ ID NOs: 10-13. In some embodiments, VL domain comprises light chain CDR1 (LCDR1) sequence comprising a sequence selected from SEQ ID NOs: 14-19, light chain CDR2 (LCDR2) sequence comprising a sequence selected from SEQ ID NOs: 20-23, and light chain CDR3 (LCDR3) sequence comprising a sequence selected from SEQ ID NOs: 24-29.

[0055] In some embodiments, the VH region of the anti-tau antibody comprises HCDR1, HCDR2, and HCDR3 sequences selected from Table 1.

TABLE-US-00001 TABLE 1 HCDR Amino Acid Sequences SEQ ID NO: HCDR1
Sequence 1 SQKVG 2 SYAMI 3 NYKVG 4 NYAMS 5 THAMT SEQ ID NO:
HCDR2 Sequence 6 IINNYGSTYYASWAKG 7 FISRSGITYYASWAKG 8
IINYYSQTTYASWAKG 9 VINPSGSAYYATWVNG SEQ ID NO: HCDR3 Sequence 10
DPDGSIVFDI 11 EFGAVGSDYYRDAFNL 12 EFGAVGSDYYRDALRL 13
DYITAGDYMDAFDP

[0056] In some embodiments, the VH region comprises HCDR1 sequence comprising SEQ ID NO: 1; HCDR2 sequence comprising SEQ ID NO: 6; and HCDR3 sequence comprising SEQ ID NO: 10. In some embodiments, the VH region comprises HCDR1 sequence comprising SEQ ID NO: 2; HCDR2 sequence comprising SEQ ID NO: 7; and HCDR3 sequence comprising SEQ ID NO: 11. In some embodiments, the VH region comprises HCDR1 sequence comprising SEQ ID NO: 3; HCDR2 sequence comprising SEQ ID NO: 8; and HCDR3 sequence comprising SEQ ID NO: 10. In some embodiments, the VH region comprises HCDR1 sequence comprising SEQ ID NO: 4; HCDR2 sequence comprising SEQ ID NO: 7; and HCDR3 sequence comprising SEQ ID NO: 12. In some embodiments, the VH region comprises HCDR1 sequence comprising SEQ ID NO: 5; HCDR2 sequence comprising SEQ ID NO: 9; and HCDR3 sequence comprising SEQ ID NO: 13.

[0057] In some embodiments, the VL region of the anti-tau antibody comprises LCDR1, LCDR2, and LCDR3 sequences selected from Table 2.

TABLE-US-00002 TABLE 2 LCDR Amino Acid Sequences SEQ ID NO: LCDR1
Sequence 14 QSSQSVVYNNRLS 15 QASESINSWLS 16 QASQNIYSNLA 17
QSSQSVYSNKRLA 18 QASQSIGSNLA 19 QASQSISNQLS SEQ ID NO: LCDR2
Sequence 20 GASTLAS 21 RASTLAS 22 GASNLAS 23 GASTLES SEQ ID NO: LCDR3
Sequence 24 LGSYDCSSGDCHA 25 QSYYEEDGIGYA 26 QGYDYSTAGAYP 27
AGGYDCSTGDCWT 28 QSYYEGSDIGYA 29 QQGYNRDNDNL

[0058] In some embodiments, the VL region comprises LCDR1 sequence comprising SEQ ID NO: 14; LCDR2 sequence comprising SEQ ID NO: 20; and LCDR3 sequence comprising SEQ ID NO: 24. In some embodiments, the VL region comprises LCDR1 sequence comprising SEQ ID NO: 15;

LCDR2 sequence comprising SEQ ID NO: 21; and LCDR3 sequence comprising SEQ ID NO: 25. In some embodiments, the VL region comprises LCDR1 sequence comprising SEQ ID NO: 16; LCDR2 sequence comprising SEQ ID NO: 22; and LCDR3 sequence comprising SEQ ID NO: 26. In some embodiments, the VL region comprises LCDR1 sequence comprising SEQ ID NO: 17; LCDR2 sequence comprising SEQ ID NO: 20; and LCDR3 sequence comprising SEQ ID NO: 27. In some embodiments, the VL region comprises LCDR1 sequence comprising SEQ ID NO: 18; LCDR2 sequence comprising SEQ ID NO: 23; and LCDR3 sequence comprising SEQ ID NO: 28. In some embodiments, the VL region comprises LCDR1 sequence comprising SEQ ID NO: 19; LCDR2 sequence comprising SEQ ID NO: 21; and LCDR3 sequence comprising SEQ ID NO: 29. [0059] In some embodiments, the anti-tau antibody is an antigen binding fragment thereof. In some embodiments, the anti-tau antibody is a chimeric antibody or antigen binding fragment thereof. In some embodiments, the anti-tau antibody comprises an IgG-scFv, nanobody, mini-antibody, minibody, scFv-CH3 KIH, Fab-scFv-Fc KIH, Fab-scFv, scFv-CH-CL-scFv, Fab', F(ab')₂, F(ab')₃, F(ab')₂-scFv₂, scFv, scFv-KIH, Fab-scFv-Fc, or intrabody. In some embodiments, the anti-tau antibody comprises a bispecific antibody. In some embodiments, the anti-tau antibody comprises a multispecific antibody. In some embodiments, the anti-tau antibody is an IgG1 antibody. In some embodiments, the anti-tau antibody is an IgG2 antibody. In some embodiments, the anti-tau antibody is an IgG4 antibody. In some embodiments, the anti-tau antibody comprises a light chain wherein the light chain is a kappa chain.

[0060] In some embodiments, the anti-tau antibody has a binding affinity to human tau of about 100 pM to about 3 nM. In some embodiments, the anti-tau antibody has a binding affinity to human tau of about 100 pM to 300 pM. In some embodiments, the anti-tau antibody has a binding affinity to human tau of about 100 pM to 500 pM. In some embodiments, the anti-tau antibody has a binding affinity to human tau of about 100 pM to 800 pM. In some embodiments, the anti-tau antibody has a binding affinity to human tau of about 300 pM to 600 pM. In some embodiments, the anti-tau antibody has a binding affinity to human tau of about 300 pM to 900 pM. In some embodiments, the anti-tau antibody has a binding affinity to human tau of about 400 pM to 1 nM. In some embodiments, the anti-tau antibody has a binding affinity to human tau of about 500 pM to 1.5 nM. In some embodiments, the anti-tau antibody has a binding affinity to human tau of about 500 pM to 2 nM. In some embodiments, the anti-tau antibody has a binding affinity to human tau of about 600 pM to 3 nM. In some embodiments, the anti-tau antibody has a binding affinity to human tau of about 100 pM to about 3 nM.

[0061] In some embodiments, the anti-tau antibody has a binding affinity to phosphorylated human tau of about 100 pM to 300 pM. In some embodiments, the anti-tau antibody has a binding affinity to phosphorylated human tau of about 100 pM to 500 pM. In some embodiments, the anti-tau antibody has a binding affinity to phosphorylated human tau of about 100 pM to 800 pM. In some embodiments, the anti-tau antibody has a binding affinity to phosphorylated human tau of about 300 pM to 600 pM. In some embodiments, the anti-tau antibody has a binding affinity to phosphorylated human tau of about 300 pM to 900 pM. In some embodiments, the anti-tau antibody has a binding affinity to phosphorylated human tau of about 400 pM to 1 nM. In some embodiments, the anti-tau antibody has a binding affinity to phosphorylated human tau of about 500 pM to 1.5 nM. In some embodiments, the anti-tau antibody has a binding affinity to phosphorylated human tau of about 500 pM to 2 nM. In some embodiments, the anti-tau antibody has a binding affinity to phosphorylated human tau of about 600 pM to 3 nM.

[0062] Described herein are antibodies comprising a sequence of any sequence set forth in Table 3 or Table 4.

TABLE-US-00003 TABLE 3 Variable Domain, Heavy Chain SEQ ID Name NO: Amino Acid Sequence Antibody 30 METGLRWLLLVAVLKGVCQSLEESGGRLVT 1 PGTPLTLTCTVSGFSLSSQKVGWVRQAPGKG Variable LEWIGIINNYGSTYYASWAKGRFTISKTTSTT Domain,

VDLRLTADLTADGCLVDPDGSIVFDIW Heavy GPGTLVTVSL Chain Antibody 31
 METGLRWLLLVAVLKGVCQCSVEESGGRLVT 2
 PGTPLTLTCTVSGFSLSSYAMIWVRQAPGKG and
 LEWIGFISRSITYYASWAKGRFTISKSTT Antibody
 VDLKMTSLTTEDTATYFCAREFGAVGSDYYR 3 DAFNLWGPGLVTVSS Variable Domain,
 Heavy Chain Antibody 32 METGLRWLLLVAVLKGVCQCSLEESGGRLVT 4
 PGTPLTLTCTVSGFSLNNYKVGWVRQAPGKG Variable
 LEWIGIINYYSQTYYASWAKGRFTISKSTT Domain,
 VDLKLTSPTTEDTATYFCARDPDGSIVFDIW Heavy GPGTLVTVSL Chain Antibody 33
 METGLRWLLLVAVLKGVCQCSVEESGGGLVT 5
 PGGTLTLTCTVSGFSLSNYAMSWVRQAPGKG Variable
 LEWIGFISRSITYYASWAKGRFTISKSTT Domain,
 VDLKITSPTTETAAYFCAREFGAVGSDYYR Heavy DALRLWGPGLVTVSS Chain
 Antibody 34 METGLRWLLLVAVLKGVCQCSLEESGGRLVT 6
 PGTPLTLTCTVSGIDLSTHAMTWVRQAPGKG Variable
 LEWIGVINPSGSAYYATWVNGRFTISKSTT Domain,
 VDLKITSPTTGDTAKYFCARDYITAGDYMD Heavy AFDPWGPGLVTVSS Chain
 TABLE-US-00004 TABLE 4 Variable Domain, Light Chain SEQ ID Name NO: Amino
 Acid Sequence Antibody 35 MDTRAPTQLLGLLLLWLPGATFAQVLTQTA 1
 SPVSAAVGGTVTINCQSSQSVVYNNRLSWF Variable
 QKPGQPPKLLIYGASTLASGVPSRFGSGS Domain,
 SGTQFTLTISDVQCDDAATYYCLGSYDCSS Light GDCHAFFGGGTEVVVK Chain Antibody
 36 MDMRAPTQLLGLLLLWLPGARCADIVMTQT 2
 PASVEAAVGGTVTINCQASESINSWLSWYQ Variable
 QKPGQPPNLLIYRASTLASGVPSRFGSGGS Domain,
 GTEYTLTISDLECADAVTYYCQSYEEDGI Light GYAFFGGGTEVVVE Chain Antibody 37
 MDMRAPTQLLGLLLLWLPGARCADIVMTQT 3
 PSSVSAAVGGTVTINCQASQNIYSNLAWYQ Variable
 QKPGQRPRLLIYGASNLASGVPSRFGSGRS Domain,
 GTEFTLTISDLECADAAATYYCQGYDYSTAG Light AYPFGGGTAVVVVK Chain Antibody 38
 MDTRAPTQLLGLLLLWLPGATFAQVLTQTA 4 SPVSAAVGSTVTINCQSSQSVYSNKRLAWF
 Variable QKPGQPPKLLIYGASTLASGVPSRFGSGS Domain,
 SGTQFTLTISDVQCDDAATYYCAGGYDCST Light GDCWTFGGGTEVVVT Chain Antibody
 39 MDMRAPTQLLGLLLLWLPGARCADIVMTQT 5
 PSSVSAAVGGTVTIKCQASQSISNLAWYQ Variable
 QKPGQPPKLLIYGASTLESQVPSRFGSGS Domain,
 GTEYTLTISDLECADAAATYYCQSYEGSDI Light GYAFFGGGTEVVVE Chain Antibody 40
 MDTRAPTQLLGLLLLWLPGARCADIVMTQT 6
 PASVSAAVGGTVTIKCQASQSISNQLSWYQ Variable
 QKSGQPPKLLIYRASTLASGVPSRFGSGS Domain,
 GTEFTLTISDLECADAAATYYCQQGYNRDNV Light DNLFGGGTEVVVK Chain

[0063] In some embodiments, the variable domain, heavy chain region (VH) comprises an amino acid sequence that has at least 70% sequence identity to the amino acid sequence according to any one of SEQ ID NOs: 30-34. In some embodiments, the VH comprises an amino acid sequence that has at least 80% sequence identity to the amino acid sequence according to any one of SEQ ID NOs: 30-34. In some embodiments, the VH comprises an amino acid sequence that has at least 85% sequence identity to the amino acid sequence according to any one of SEQ ID NOs: 30-34. In some embodiments, the VH comprises an amino acid sequence that has at least 90% sequence identity to the amino acid sequence according to any one of SEQ ID NOs: 30-34. In some embodiments, the VH comprises an amino acid sequence that has at least 91% sequence identity to

and the VL comprises an amino acid sequence that has at least 94% sequence identity to the amino acid sequence according to SEQ ID NO: 36. In some embodiments, the VH comprises an amino acid sequence that has at least 95% sequence identity to the amino acid sequence according to SEQ ID NO: 31; and the VL comprises an amino acid sequence that has at least 95% sequence identity to the amino acid sequence according to SEQ ID NO: 36. In some embodiments, the VH comprises an amino acid sequence that has at least 96% sequence identity to the amino acid sequence according to SEQ ID NO: 31; and the VL comprises an amino acid sequence that has at least 96% sequence identity to the amino acid sequence according to SEQ ID NO: 36. In some embodiments, the VH comprises an amino acid sequence that has at least 97% sequence identity to the amino acid sequence according to SEQ ID NO: 31; and the VL comprises an amino acid sequence that has at least 97% sequence identity to the amino acid sequence according to SEQ ID NO: 36. In some embodiments, the VH comprises an amino acid sequence that has at least 98% sequence identity to the amino acid sequence according to SEQ ID NO: 31; and the VL comprises an amino acid sequence that has at least 98% sequence identity to the amino acid sequence according to SEQ ID NO: 36. In some embodiments, the VH comprises an amino acid sequence that has at least 99% sequence identity to the amino acid sequence according to SEQ ID NO: 31; and the VL comprises an amino acid sequence that has at least 99% sequence identity to the amino acid sequence according to SEQ ID NO: 36.

[illegible]

acid sequence according to SEQ ID NO: 40. In some embodiments, the VH comprises an amino acid sequence that has at least 85% sequence identity to the amino acid sequence according to SEQ ID NO: 34; and the VL comprises an amino acid sequence that has at least 85% sequence identity to the amino acid sequence according to SEQ ID NO: 40. In some embodiments, the VH comprises an amino acid sequence that has at least 90% sequence identity to the amino acid sequence according to SEQ ID NO: 34; and the VL comprises an amino acid sequence that has at least 90% sequence identity to the amino acid sequence according to SEQ ID NO: 40. In some embodiments, the VH comprises an amino acid sequence that has at least 91% sequence identity to the amino acid sequence according to SEQ ID NO: 34; and the VL comprises an amino acid sequence that has at least 91% sequence identity to the amino acid sequence according to SEQ ID NO: 40. In some embodiments, the VH comprises an amino acid sequence that has at least 92% sequence identity to the amino acid sequence according to SEQ ID NO: 34; and the VL comprises an amino acid sequence that has at least 92% sequence identity to the amino acid sequence according to SEQ ID NO: 40. In some embodiments, the VH comprises an amino acid sequence that has at least 93% sequence identity to the amino acid sequence according to SEQ ID NO: 34; and the VL comprises an amino acid sequence that has at least 93% sequence identity to the amino acid sequence according to SEQ ID NO: 40. In some embodiments, the VH comprises an amino acid sequence that has at least 94% sequence identity to the amino acid sequence according to SEQ ID NO: 34; and the VL comprises an amino acid sequence that has at least 94% sequence identity to the amino acid sequence according to SEQ ID NO: 40. In some embodiments, the VH comprises an amino acid sequence that has at least 95% sequence identity to the amino acid sequence according to SEQ ID NO: 34; and the VL comprises an amino acid sequence that has at least 95% sequence identity to the amino acid sequence according to SEQ ID NO: 40. In some embodiments, the VH comprises an amino acid sequence that has at least 96% sequence identity to the amino acid sequence according to SEQ ID NO: 34; and the VL comprises an amino acid sequence that has at least 96% sequence identity to the amino acid sequence according to SEQ ID NO: 40. In some embodiments, the VH comprises an amino acid sequence that has at least 97% sequence identity to the amino acid sequence according to SEQ ID NO: 34; and the VL comprises an amino acid sequence that has at least 97% sequence identity to the amino acid sequence according to SEQ ID NO: 40. In some embodiments, the VH comprises an amino acid sequence that has at least 98% sequence identity to the amino acid sequence according to SEQ ID NO: 34; and the VL comprises an amino acid sequence that has at least 98% sequence identity to the amino acid sequence according to SEQ ID NO: 40. In some embodiments, the VH comprises an amino acid sequence that has at least 99% sequence identity to the amino acid sequence according to SEQ ID NO: 34; and the VL comprises an amino acid sequence that has at least 99% sequence identity to the amino acid sequence according to SEQ ID NO: 40.

[0081] Described herein, in some embodiments, are antibodies or antibody fragments comprising a heavy chain sequence at least about 90% identical to a sequence as set forth in any one of SEQ ID NOs: 41, 43, 46, 48, and 50. In some instances, the antibodies or antibody fragments comprise a heavy chain sequence at least or about 70%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity to any one of SEQ ID NOs: 41, 43, 46, 48, and 50.

[0082] Described herein, in some embodiments, are antibodies or antibody fragments comprising a light chain sequence at least about 90% identical to a sequence as set forth in any one of SEQ ID NOs: 42, 44, 45, 47, 49, and 51. In some instances, the antibodies or antibody fragments comprise a light chain sequence at least or about 70%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity to any one of SEQ ID NOs: 42, 44, 45, 47, 49, and 51.

[0083] Described herein, in some embodiments, are antibodies or antibody fragments comprising a heavy chain sequence at least about 90% identical to a sequence as set forth in any one of SEQ ID NOs: 41, 43, 46, 48, and 50 and a light chain sequence at least about 90% identical to a sequence as set forth in any one of SEQ ID NOs: 42, 44, 45, 47, 49, and 51. In some instances, the antibodies or

antibody fragments comprise a heavy chain sequence at least or about 70%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity to any one of SEQ ID NOs: 41, 43, 46, 48, and 50 and a light chain sequence at least or about 70%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity to any one of SEQ ID NOs: 42, 44, 45, 47, 49, and 51.

TABLE-US-00005 TABLE 5 Heavy Chain and Light Chain Sequences SEQ ID Name
NO: Amino Acid Sequence Antibody 41

METGLRWLLLVAVLKGVCQSLEESGGRLVTPGTPLTLTCTVSG 1
FSLSSQKVGWVRQAPGKGLEWIGIINNYGSTYYASWAKGRFTIS Heavy
KTSTTVDLRITSLTAEDTATYFCARPDGSIWFDIWGPGLTVTVSL Chain
GQPKAPSVFPLAPCCGDTPSSTVTGCLVKGYLPEPVTVTWNSG
TLTNGVRTFPSVRQSSGLYSLSSVSVTSSSQPVTCNV AHPATNT
KVDKTVAPSTCSKPTCPPPELLGRSSVFIFPPKPKDTLMISRTPEV
TCVVVDVSQDDPEVQFTWYINNEQVRTARPPPLREQQFNSTIRVV
STLPIAHQDWLRGKEFKCKVHNKALPAPIEKTISKARGQPLEPKV
YTMGPPREELSSRSVSLTCMINGFYPSDISVEWEKNGKAEDNYK
TTPAVLDSGYSFLYSKLSVPTSEWQRGDVFTCSVMHEALHNH YTQKSISRSPGK Antibody

42 MDTRAPTQLLGLLLLWLPGATFAQVLTQTASPVSAAVGGTVTI 1
NCQSSQSVVYNNRLSWFQQKPGQPPKLLIYGASTLASGVPSRF Light
KGSQSGTQFTLTISDVQCDDAATYYCLGSYDCSSGDCHAFFGGG Chain
TEVVVKGDVPVAPTFLIFPPAADQVATGTVTIVCVANKYFPDVT
VTWEVDGTTQTTGIENSKTPQNSADCTYNLSSTLTLTSTQYNH
KEYTCKVTQGTTSVVQSFNRGDC Antibody 43

METGLRWLLLVAVLKGVCQSVEESGGRLVTPGTPLTLTCTVSG 2
FSLSSYAMIWVRQAPGKGLEWIGIFISRSGITYYASWAKGRFTISK and
TSTTVDLKMTSLTTEDTATYFCAREFGAVGSDYYRDAFNLWGP Antibody
GTLTVTVSSGQPKAPSVFPLAPCCGDTPSSTVTGCLVKGYLPEPV 3
TVTWNSTLTNGVRTFPSVRQSSGLYSLSSVSVTSSSQPVTCNV Heavy
AHPATNTKVDKTVAPSTCSKPTCPPPELLGRSSVFIFPPKPKDTL Chain
MISRTPEVTCVVVDVSQDDPEVQFTWYINNEQVRTARPPPLREQQ
FNSTIRVVSTLPIAHQDWLRGKEFKCKVHNKALPAPIEKTISKAR
GQPLEPKVYTMGPPREELSSRSVSLTCMINGFYPSDISVEWEKNG
KAEDNYKTTPAVLDSGYSFLYSKLSVPTSEWQRGDVFTCSVM
HEALHNHYTQKSISRSPGK Antibody 44

MDMRAPTQLLGLLLLWLPGARCADIVMTQTPASVEAAVGGTV 2
TINCQASESINSWLSWYQQKPGQPPNLLIYRASTLASGVPSRFSG Light
GGSGTEYTLTISDLECAVATYYCQSYEEDGIGYAFGGGTEVV Chain
VEGDPVAPTFLIFPPAADQVATGTVTIVCVANKYFPDVTVTWE
VDGTTQTTGIENSKTPQNSADCTYNLSSTLTLTSTQYNHKEYT
CKVTQGTTSVVQSFNRGDC Antibody 45

MDMRAPTQLLGLLLLWLPGARCADIVMTQTPSSVSAVGGTVT 3
INCQASQNIYSNLAWYQQKPGQRPRLIYGASNLASGVPSRFKG Light
SRSGTEFTLTISDLECAATYYCQGYDYTAGAYPFGGGTAVV Chain
VKGDPVAPTFLIFPPAADQVATGTVTIVCVANKYFPDVTVTWEV
DGTTQTTGIENSKTPQNSADCTYNLSSTLTLTSTQYNHKEYTCK
VTQGTTSVVQSFNRGDC Antibody 46

METGLRWLLLVAVLKGVCQSLEESGGRLVTPGTPLTLTCTVS 4
GFSLNKYKVGWVRQAPGKGLEWIGIINYYSQTYASWAKGRF Heavy
TISKSTTVDLKLTSTTTEDTATYFCARPDGSIWFDIWGPGLTV Chain
TVSLGQPKAPSVFPLAPCCGDTPSSTVTGCLVKGYLPEPVTVT

WNSGTLTNGVLSNGLSLSSVVSVTSSSQPVTCNVAH
PATNTKVDKTVVPSTCSKPTCPPPELLGRSSVFIFPPKPKDTLMI
SRTPEVTCVVVDVSQDDPEVQFTWYINNEQVRTARPPLREQQF
NSTIRVVSTLPIAHQDWLRGKEFKCKVHNKALPAPIEKTISKAR
GQPLEPKVYTMGPPREELSSRSVSLTCMINGFYPSDISVEWEKN
GKAEDNYKTTPAVLDSGSGYFLYSKLSVPTSEWQRGDVFTCSV
MHEALHNHYTQKSISRSPGK Antibody 47
MDTRAPTQLLGLLLLWLPGATFAQVLTQTASPVSAAVGSTVTIN 4
CQSSQSVYSNKRLLAWFQLKPGQPPKLLIYGASTLASGVPSRFGK Light
SGSGTQFTLTISDVQCDDAATYYCAGGYDCSTGDCWTFGGGTE Chain
VVVTGDPVAPTVLIFPPAADQVATGTVTIVCVANKYFPDVTVT
WEVDGTTQTTGIENSKTPQNSADCTYNLSSTLTLTSTQYNHKE
YTCKVTQGTTSVVQSFNRGDC Antibody 48
METGLRWLLLVAVLKGVQCQSVEESGGGLVTPGGTLTLTCTVS 5
GFSLSNYAMSWVRQAPGKGLEWIGFISRSITYYASWAKGRFT Heavy
ISKSTTTVDLKITSPPTEDTAAYFCAREFGAVGSDYYRDALRLW Chain
GPGTLVTVSSGQPKAPSVFPLAPCCGDTPTSSTVTLGCLVKGYLP
EPVTVTWNSGTLTNGVVRTFPSVRQSSGLYSLSSVVSVTSSSQPV
CNVAHPATNTKVDKTVAPSTCSKPTCPPPELLGRSSVFIFPPKPK
DTLMISRTPEVTCVVVDVSQDDPEVQFTWYINNEQVRTARPPL
REQQFNSTIRVVSTLPIAHQDWLRGKEFKCKVHNKALPAPIEKT
ISKARGQPLEPKVYTMGPPREELSSRSVSLTCMINGFYPSDISVE
WEKNGKAEDNYKTTPAVLDSGSGYFLYSKLSVPTSEWQRGDV
FTCSVMHEALHNHYTQKSISRSPGK Antibody 49
MDMRAPTQLLGLLLLWLPGARCADIVMTQTPSSVSAAVGGTVT 5
IKCQASQSISNLAWEYQQKPGQPPKLLIYGASTLESGVPSRFGKS Light
GSGTEYTLTISDLECADAAATYYCQSYEYEGSDIGYAFGGGTEVVV Chain
EGDPVAPTVLIFPPAADQVATGTVTIVCVANKYFPDVTVTWEVD
GTTQTTGIENSKTPQNSADCTYNLSSTLTLTSTQYNHKEYTCKV TQGTTSVVQSFNRGDC
Antibody 50 METGLRWLLLVAVLKGVQCQSLEESGGRLVTPGTPLTLTCTVS 6
GIDLSTHAMTWVRQAPGKGLEWIGVINPSGSAYYATWVNGRF Heavy
TISKSTTTVDLKITSPTTGDTAKYFCARDYITAGDYMDAFDPW Chain
GPGTLVTVSSGQPKAPSVFPLAPCCGDTPTSSTVTLGCLVKGYLP
EPVTVTWNSGTLTNGVVRTFPSVRQSSGLYSLSSVVSVTSSSQPV
TCNVAHPATNTKVDKTVAPSTCSKPTCPPPELLGRSSVFIFPPKPK
KDTLMISRTPEVTCVVVDVSQDDPEVQFTWYINNEQVRTARPP
LREQQFNSTIRVVSTLPIAHQDWLRGKEFKCKVHNKALPAPIEK
TISKARGQPLEPKVYTMGPPREELSSRSVSLTCMINGFYPSDISV
EWEKNGKAEDNYKTTPAVLDSGSGYFLYSKLSVPTSEWQRGD
VFTCSVMHEALHNHYTQKSISRSPGK Antibody 51
MDTRAPTQLLGLLLLWLPGARCADIVMTQTPASVSAAVGGTV 6
TIKCQASQSISNQLSWYQQKSGQPPKLLIYRASTLASGVPSRFK Light
GSGSGTEFTLTISDLECADAAATYYCQQGYNRDNVDNLFGGGT Chain
EVVVKGDPVAPTVLIFPPAADQVATGTVTIVCVANKYFPDVTV
TWEVDGTTQTTGIENSKTPQNSADCTYNLSSTLTLTSTQYNH
KEYTCKVTQGTTSVVQSFNRGDC

[0084] In some embodiments, the anti-tau antibody comprises a VH domain that is encoded by a nucleic acid comprising at least 80%, at least 85%, at least 90%, at least 95% sequence identity to a sequence selected from SEQ ID NOs: 52-56. In some embodiments, the anti-tau antibody comprises a VL domain that is encoded by a nucleic acid comprising at least 80%, at least 85%, at

[illegible]

comprising a sequence identical to SEQ ID NO: 58. In some embodiments, the anti-tau antibody comprises a VL domain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NO: 59. In some embodiments, the anti-tau antibody comprises a VL domain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NO: 60. In some embodiments, the anti-tau antibody comprises a VL domain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NO: 61. In some embodiments, the anti-tau antibody comprises a VL domain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NO: 62. In some embodiments, the anti-tau antibody comprises a VH domain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NO: 52 and a VL domain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NO: 57. In some embodiments, the anti-tau antibody comprises a VH domain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NO: 53 and a VL domain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NO: 58. In some embodiments, the anti-tau antibody comprises a VH domain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NO: 53 and a VL domain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NO: 59. In some embodiments, the anti-tau antibody comprises a VH domain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NO: 54 and a VL domain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NO: 60. In some embodiments, the anti-tau antibody comprises a VH domain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NO: 55 and a VL domain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NO: 61. In some embodiments, the anti-tau antibody comprises a VH domain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NO: 56 and a VL domain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NO: 62.

TABLE-US-00006 TABLE 6 Nucleic acid sequences encoding VH domains SEQ ID NO: Nucleic acid sequences encoding VH domains 52

ATGGAGACTGGGCTGCGCTGGCTTCTCCTGGTCGCTGTGCTCAA
AGGTGTCCAGTGTCAGTCGCTGGAGGAGTCCGGGGGTCGCCTGG
TCACGCCTGGGACACCCCTGACACTCACCTGCACAGTCTCTGGA
TTTTCCCTCAGTAGCCAGAAAGTGGGCTGGGTCCGCCAGGCTCC
AGGGAAGGGGCTGGAATGGATCGGAATCATTAATAATTATGGTA
GCACATACTACGCGAGCTGGGCGAAAGGCCGATTCACCATCTCG
AAAACCTCGACCACAGTGGATCTGAGAATCACCAGTCTGACGGC
CGAGGACACGGCCACCTATTTCTGTGCCCCGTGATCCTGATGGTA
GTATTGTCTTTGACATCTGGGGCCCAGGCACCCTTGTCACCGTC TCCTTG 53

ATGGAGACTGGGCTGCGCTGGCTTCTCCTGGTCGCTGTGCTCAA
AGGTGTCCAGTGTCAGTCGGTGGAGGAGTCCGGGGGTCGCCTGG
TCACGCCTGGGACACCCCTGACACTCACCTGCACCGTCTCTGGA
TTCTCCCTCAGTAGCTATGCAATGATCTGGGTCCGCCAGGCTCC
AGGGAAGGGGCTGGAATGGATCGGATTCATTAGTCGTAGTGGTA
TCACATACTACGCGAGCTGGGCAAAGGCCGATTCACCATCTCC
AAAACCTCGACCACGGTGGATCTGAAAATGACCAGTCTGACAAC
CGAGGACACGGCCACCTATTTCTGTGCCAGAGAATTCGGTGCTG
TTGGTAGTGATTATTATAGGGACGCCTTTAACTTGTGGGGCCCA
GGCACCTGGTCACCGTCTCCTCA 54

ATGGAGACTGGGCTGCGCTGGCTTCTCCTGGTCGCTGTGCTCAA
AGGTGTCCAGTGTCAGTCGCTGGAGGAGTCCGGGGGTCGCCTGG
TCACGCCTGGGACACCCCTGACACTCACCTGCACAGTCTCTGGA
TTTTCCCTAAATAACTACAAAGTGGGCTGGGTCCGCCAGGCTCC
AGGAAAGGGGCTGGAATGGATCGGAATCATTAATAATTATAGTC
AGACATACTACGCGAGCTGGGCAAAGGCCGATTCACCATCTCG

AAAACCTCGACCTCGTCAAGCTCAGCTCACCAGTCCGACAAC
CGAAGACACGGCCACCTATTTCTGTGCCCCGTGATCCTGATGGTA
GTATTGTCTTTGACATCTGGGGCCCAGGCACCCTTGTCACCGTC TCCTTG 55
ATGGAGACTGGGCTGCGCTGGCTTCTCCTGGTCGCTGTGCTCAA
AGGTGTCCAGTGTCAGTCGGTGGAGGAGTCCGGAGGAGGCCTGG
TAACGCCTGGAGGAACCCTGACACTCACCTGCACCGTCTCTGGA
TTCTCCCTCAGTAACTATGCAATGAGCTGGGTCCGCCAGGCTCC
AGGGAAGGGGCTGGAATGGATCGGATTCATTAGTCGTAGTGGTA
TTACATACTACGCGAGCTGGGCAAAAGGCCGATTCACCATCTCC
AAAACCTCGACCACGGTGGATCTGAAAATCACCAGTCCGACGAC
CGAGGACACGGCCGCCTATTTCTGTGCCAGAGAATTCCGGTGCTG
TTGGTAGTGATTATTATAGGGACGCCTTGAGGTTGTGGGGCCCA
GGCACCCCTGGTCACCGTCTCCTCA 56

ATGGAGACTGGGCTGCGCTGGCTTCTCCTGGTCGCTGTGCTCAA
AGGTGTCCAGTGTCAGTCGCTGGAGGAGTCCGGGGGTGCGCTGG
TAACGCCTGGGACACCCCTGACACTCACCTGCACAGTCTCTGGA
ATCGACCTCAGTACCCATGCAATGACCTGGGTCCGCCAGGCTCC
AGGAAAGGGGCTGGAATGGATCGGAGTCATTAATCCTAGTGGTA
GCGCATACTACGCGACCTGGGTGAATGGCCGATTCACCATCTCC
AAAACCTCGACCACGGTGGATCTGAAAATCACCAGTCCGACAAC
CGGGGACACGGCCAAGTATTTCTGTGCCAGAGATTATATTACTG
CGGGTGATTATTATATGGATGCTTTTGATCCCTGGGGCCCAGGC
ACCCTGGTCACCGTCTCCTCA

TABLE-US-00007 TABLE 7 Nucleic acid sequences encoding VL domains SEQ ID
NO: Nucleic acid sequences encoding VL domains 57

ATGGACACGAGGGCCCCCACTCAGCTGCTGGGGCTCCTGCTGCT
CTGGCTCCCAGGTGCCACATTTGCCCAAGTGCTGACCCAGACTG
CATCCCCCGTGTCTGCGGCTGTTGGAGGCACAGTCACCATCAAT
TGCCAGTCCAGTCAGAGTGTTGTATATAACAACCGCTTATCCTG
GTTTCAACAGAAACCAGGGCAGCCTCCCAAGCTCCTGATCTATG
GTGCATCCACTCTGGCATCTGGGGTCCCATCGCGGTTCAAAGGC
AGTGGATCTGGGACACAGTTCCTCTCACCATCAGCGACGTGCA
GTGTGACGATGCTGCCACTTACTACTGTCTAGGCTCCTATGATT
GTAGTAGTGGTGATTGCCATGCTTTTCGGCGGAGGGACCGAGGTG GTGGTCAA 58

ATGGACATGAGGGCCCCCACTCAGCTGCTGGGGCTCCTACTGCT
CTGGCTCCCAGGTGCCAGATGTGCTGACATTGTGATGACCCAGA
CTCCAGCCTCCGTGGAGGCAGCTGTGGGAGGCACAGTCACCATC
AATTGCCAAGCCAGTGAGAGCATTAAATAGTTGGTTGTCCTGGTA
TCAGCAGAAACCAGGGCAGCCTCCCAACCTCCTGATCTACAGGG
CATCCACTCTGGCATCTGGGGTCCCATCGCGGTTCAAGTGGCGGT
GGATCTGGGACAGAGTACACTCTCACCATCAGCGACCTGGAGTG
TGCCGATGCTGTCATTATTACTGTCAAAGCTATTATGAGGAGG
ATGGTATTGGTTATGCTTTTCGGCGGAGGGACCGAGGTGGTGGTC GAA 59
ATGGACATGAGGGCCCCCACTCAGCTGCTGGGGCTCCTGCTGCT
CTGGCTCCCAGGTGCCAGATGTGCTGACATTGTGATGACCCAGA
CTCCATCCTCCGTGTCTGCAGCTGTGGGAGGCACAGTCACCATC
AATTGCCAGGCCAGTCAGAACATTTACAGCAATTTAGCCTGGTA
TCAGCAGAAACCAGGGCAGCGTCCCAGGCTCCTGATCTATGGCG
CATCCAATCTGGCATCTGGGGTCCCATCGCGGTTCAAAGGCAGT
AGATCTGGGACAGAGTTCCTCTCACCATCAGCGACCTGGAGTG

TGCCGATGCTGCCACTTACTGTCAAGGCTATGATTATAGTA
 CTGCTGGTGCCTATCCTTTTCGGCGGAGGGACCGCGGTGGTGGTC AAA 60
 ATGGACACGAGGGCCCCCACTCAGCTGCTGGGGCTCCTGCTGCT
 CTGGCTCCCAGGTGCCACATTTGCCCAAGTGCTGACCCAGACTG
 CATCGCCCGTGTCTGCGGCTGTGGGAAGCACAGTCACCATCAAT
 TGCCAGTCCAGTCAGAGCGTTTATAGTAACAAGCGCTTAGCCTG
 GTTTCAGCTGAAACCAGGGCAGCCTCCCAAGCTCCTGATCTATG
 GTGCATCCACACTGGCATCTGGGGTCCCATCGCGATTCAAGGGC
 AGTGGATCTGGGACACAGTTCCTCCTCACCATCAGCGACGTGCA
 GTGTGACGATGCTGCCACTTACTACTGTGCAGGCGGTTATGATT
 GTAGTACTGGTGATTGTTGGACTTTTCGGCGGAGGGACCGAGGTG GTGGTCACA 61
 ATGGACATGAGGGCCCCCACTCAGCTGCTGGGGCTCCTGCTGCT
 CTGGCTCCCAGGTGCCAGATGTGCTGACATCGTGATGACCCAGA
 CTCCATCCTCCGTGTCTGCAGCTGTGGGAGGCACAGTCACCATC
 AAGTGCCAGGCCAGTCAGAGCATTGGTAGTAATTTAGCCTGGTA
 TCAGCAGAAACCAGGGCAGCCTCCCAAGCTCCTGATCTATGGTG
 CATCCACTCTGGAATCTGGGGTCCCATCGCGGTTTAAAGGCAGT
 GGATCTGGGACAGAGTACACTCTCACCATCAGCGACCTGGAGTG
 TGCCGATGCTGCCACTTACTACTGTCAAAGCTATTATGAGGGTA
 GTGATATTGGTTATGCTTTTCGGCGGAGGGACCGAGGTGGTGGTC GAA 62
 ATGGACACGAGGGCCCCCACTCAGCTGCTGGGGCTCCTGCTGCT
 CTGGCTCCCAGGTGCCAGATGTGCTGACATCGTGATGACCCAGA
 CTCCAGCCTCTGTGTCTGCAGCTGTGGGAGGCACAGTCACCATC
 AAGTGCCAGGCCAGTCAGAGCATTAGCAACCAACTATCCTGGTA
 TCAGCAGAAATCAGGGCAGCCTCCCAAGCTCCTGATCTACAGGG
 CATCTACTCTGGCATCTGGGGTCCCATCGCGGTTCAAAGGCAGT
 GGATCTGGGACAGAGTTCCTCCTCACCATCAGCGACCTGGAGTG
 TGCCGATGCTGCCACTTACTACTGTCAACAGGGTTATAATAGAG
 ATAATGTTGATAATCTTTTCGGCGGAGGGACCGAGGTGGTGGTC AAA

[0085] In some embodiments, the anti-tau antibody comprises a heavy chain that is encoded by a nucleic acid comprising a sequence identical to a sequence selected from SEQ ID NOs: 63-67. In some embodiments, the anti-tau antibody comprises a light chain that is encoded by a nucleic acid comprising a sequence identical to a sequence selected from SEQ ID NOs: 68-73. Nucleic acid sequences for heavy chains for anti-tau antibodies described here are listed in Table 8 and nucleic acid sequences for light chains for anti-tau antibodies described here are listed in Table 9. Nucleic acid sequences listed in Table 8 and Table 9 may be used in the process of in vitro production of antibodies described herein. In some embodiments, the anti-tau antibody comprises a heavy chain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NO: 63. In some embodiments, the anti-tau antibody comprises a heavy chain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NO: 64. In some embodiments, the anti-tau antibody comprises a heavy chain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NO: 65. In some embodiments, the anti-tau antibody comprises a heavy chain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NO: 66. In some embodiments, the anti-tau antibody comprises a heavy chain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NO: 67. In some embodiments, the anti-tau antibody comprises a light chain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NO: 68. In some embodiments, the anti-tau antibody comprises a light chain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NO: 69. In some embodiments, the anti-tau antibody comprises a light chain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NO: 70. In some embodiments, the anti-tau antibody comprises a light chain that is encoded by

a nucleic acid comprising a sequence identical to SEQ ID NO: 71. In some embodiments, the anti-tau antibody comprises a light chain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NO: 72. In some embodiments, the anti-tau antibody comprises a light chain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NO: 73. In some embodiments, the anti-tau antibody comprises a heavy chain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NO: 63 and a light chain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NO: 68. In some embodiments, the anti-tau antibody comprises a heavy chain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NO: 64 and a light chain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NO: 69. In some embodiments, the anti-tau antibody comprises a heavy chain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NO: 64 and a light chain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NO: 70. In some embodiments, the anti-tau antibody comprises a heavy chain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NO: 65 and a light chain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NO: 71. In some embodiments, the anti-tau antibody comprises a heavy chain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NO: 66 and a light chain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NO: 72. In some embodiments, the anti-tau antibody comprises a heavy chain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NO: 67 and a light chain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NO: 73.

TABLE-US-00008 TABLE 8 Nucleic acid sequences encoding heavy chains SEQ ID NO: Nucleic acid sequences encoding heavy chains 63

ATGGAGACTGGGCTGCGCTGGCTTCTCCTGGTCGCTGTGCTCAAA
GGTGTCCAGTGTCTAGTCGCTGGAGGAGTCCGGGGGTCGCCTGGTC
ACGCCTGGGACACCCCTGACACTCACCTGCACAGTCTCTGGATTT
TCCCTCAGTAGCCAGAAAGTGGGCTGGGTCCGCCAGGCTCCAGGG
AAGGGGCTGGAATGGATCGGAATCATTAATAATTATGGTAGCACA
TACTACGCGAGCTGGGCGAAAGGCCGATTCACCATCTCGAAAACC
TCGACCACAGTGGATCTGAGAATCACAGTCTGACGGCCGAGGAC
ACGGCCACCTATTTCTGTGCCCGTGATCCTGATGGTAGTATTGTC
TTTGACATCTGGGGCCCAGGCACCCTTGTCACCGTCTCCTTGGGG
CAACCTAAGGCTCCATCAGTCTTCCCCTGGCCCCCTGCTGCGGG
GACACACCCAGCTCCACGGTGACCCTGGGCTGCCTGGTCAAAGGC
CTACTCCCGGAGCCAGTGACCGTGACCTGGAACCTCGGGCACCCCTC
ACCAATGGGGTACGCACCTTCCCGTCCGTCCGGCAGTCCTCAGGC
CTCTACTCGCTGAGCAGCGTGGTGAGCGTGACCTCAAGCAGCCAG
CCCGTCACCTGCAACGTGGCCCACCCAGCCACCAACACCAAAGTG
GACAAGACCGTTGCGCCCTCGACATGCAGCAAGCCCACGTGCCCA
CCCCCTGAACCTCCTGGGGCGATCCTCTGTCTTCATCTTCCCCCA
AAACCCAAGGACACCCTCATGATCTCACGCACCCCCGAGGTCACA
TGCGTGGTGGTGGACGTGAGCCAGGATGACCCCGAGGTGCAGTTC
ACATGGTACATAAACAACGAGCAGGTGCGCACCGCCCGGCCGCG
CTACGGGAGCAGCAGTTCAACAGCACGATCCGCGTGGTCAGCACC
CTCCCCATCGCGCACCAAGGACTGGCTGAGGGGCAAGGAGTTCAAG
TGCAAAGTCCACAACAAGGCACTCCCGGCCCCCATCGAGAAAACC
ATCTCCAAAGCCAGAGGGCAGCCCCTGGAGCCGAAGGTCTACACC
ATGGGCCCTCCCCGGGAGGAGCTGAGCAGCAGGTCCGGTCAGCCTG
ACCTGCATGATCAACGGCTTCTACCCTTCCGACATCTCGGTGGAG
TGGGAGAAGAACGGGAAGGCAGAGGACAATAAGACCACGCCG
GCCGTGCTGGACAGCGACGGCTCCTACTTCCTCTACAGCAAGCTC

TCAGTGCAGGCGGGCGGCGAGCGTCTTACCTGC
TCCGTGATGCACGAGGCCTTGCACAACCACTACACGCAGAAGTCC
ATCTCCCGCTCTCCGGGTAAATGA 64
ATGGAGACTGGGCTGCGCTGGCTTCTCCTGGTCGCTGTGCTCAAA
GGTGTCCAGTGTGAGTCGGTGGAGGAGTCCGGGGGGTCGCCTGGTC
ACGCCTGGGACACCCCTGACACTCACCTGCACCGTCTCTGGATT
TCCCTCAGTAGCTATGCAATGATCTGGGTCCGCCAGGCTCCAGGG
AAGGGGCTGGAATGGATCGGATTCATTAGTCGTAGTGGTATCACA
TACTACGCGAGCTGGGCAAAAGGCCGATTCACCATCTCCAAAACC
TCGACCACGGTGGATCTGAAAATGACCAGTCTGACAACCGAGGAC
ACGGCCACCTATTTCTGTGCCAGAGAATTCGGTGCTGTTGGTAGT
GATTATTATAGGGACGCCTTTAACTTGTGGGGGCCAGGCACCCTG
GTCACCGTCTCCTCAGGGCAACCTAAGGCTCCATCAGTCTTCCCA
CTGGCCCCCTGCTGCGGGGACACACCCAGCTCCACGGTGACCCTG
GGCTGCCTGGTCAAAGGCTACCTCCCGGAGCCAGTGACCGTGACC
TGGAACCTCGGGCACCCCTACCAATGGGGTACGCACCTTCCCGTCC
GTCCGGCAGTCCTCAGGCCTCTACTCGCTGAGCAGCGTGGTGAGC
GTGACCTCAAGCAGCCAGCCCGTCACCTGCAACGTGGCCCACCCA
GCCACCAACACCAAAGTGGACAAGACCGTTGCGCCCTCGACATGC
AGCAAGCCCACGTGCCCACCCCTGAACTCCTGGGGCGATCCTCT
GTCTTCATCTTCCCCCCTAAAACCCAAGGACACCCTCATGATCTCA
CGCACCCCGGAGGTACATGCGTGGTGGTGGACGTGAGCCAGGAT
GACCCCGAGGTGCAGTTCACATGGTACATAACAACGAGCAGGTG
CGCACCGCCCGGCCGCGCTACGGGAGCAGCAGTTCAACAGCACG
ATCCGCGTGGTCAGCACCCCTCCCCATCGCGCACCAAGGACTGGCTG
AGGGGCAAGGAGTTCAAGTGCAAAGTCCACAACAAGGCACTCCCG
GCCCCCATCGAGAAAACCATCTCCAAAGCCAGAGGGCAGCCCCTG
GAGCCGAAGGTCTACACCATGGGGCCCTCCCCGGGAGGAGCTGAGC
AGCAGGTCGGTCAGCCTGACCTGCATGATCAACGGCTTCTACCCT
TCCGACATCTCGGTGGAGTGGGAGAAGAACGGGAAGGCAGAGGAC
AACTACAAGACCACGCCGGCCGTGCTGGACAGCGACGGCTCCTAC
TTCCTCTACAGCAAGCTCTCAGTGCCACAGAGTGAGTGGCAGCGG
GGCGACGTCTTCACCTGCTCCGTGATGCACGAGGCCTTGACAAAC
CACTACACGCAGAAGTCCATCTCCCGCTCTCCGGGTAAATGA 65
ATGGAGACTGGGCTGCGCTGGCTTCTCCTGGTCGCTGTGCTCAAA
GGTGTCCAGTGTGAGTCGCTGGAGGAGTCCGGGGGGTCGCCTGGTC
ACGCCTGGGACACCCCTGACACTCACCTGCACAGTCTCTGGATTT
TCCCTAAATAACTACAAAGTGGGCTGGGTCCGCCAGGCTCCAGGA
AAGGGGCTGGAATGGATCGGAATCATTAATACTATTATAGTCAGACA
TACTACGCGAGCTGGGCCAAAGGCCGATTCACCATCTCGAAAACC
TCGACCACGGTGGATCTGAAGCTCACCAAGTCCGACAACCGAAGAC
ACGGCCACCTATTTCTGTGCCCGTGATCCTGATGGTAGTATTGTC
TTTGACATCTGGGGGCCAGGCACCCTTGTCACCGTCTCCTTGGGG
CAACCTAAGGCTCCATCAGTCTTCCCCTGGCCCCCTGCTGCGGG
GACACACCCAGCTCCACGGTGACCCTGGGCTGCCTGGTCAAAGGC
TACCTCCCGGAGCCAGTGACCGTGACCTGGAACCTCGGGCACCCCTC
ACCAATGGGGTACGCACCTTCCCGTCCGTCCGGCAGTCCTCAGGC
CTCTACTCGCTGAGCAGCGTGGTGAGCGTGACCTCAAGCAGCCAG
CCCGTCACCTGCAACGTGGGCCACCCAGCCACCAACACCAAAGTG
GACAAGACCGTTGTGCCCTCGACATGCAGCAAGCCCACGTGCCCA

CCCTCTGAACCTGGGGCTGCTCTTCATCTTCCCCCA
AAACCCAAGGACACCCTCATGATCTCACGCACCCCGAGGTCACA
TGCGTGGTGGTGGACGTGAGCCAGGATGACCCCGAGGTGCAGTTC
ACATGGTACATAAACACGAGCAGGTGCGCACCGCCCGGCCGCCG
CTACGGGAGCAGCAGTTCAACAGCACGATCCGCGTGGTCAGCACC
CTCCCCATCGCGCACCAAGGACTGGCTGAGGGGCAAGGAGTTCAAG
TGCAAAGTCCACAACAAGGCACTCCCGGCCCCCATCGAGAAAACC
ATCTCCAAAGCCAGAGGGCAGCCCCTGGAGCCGAAGGTCTACACC
ATGGGCCCTCCCCGGGAGGAGCTGAGCAGCAGGTTCGGTCAGCCTG
ACCTGCATGATCAACGGCTTCTACCCTTCCGACATCTTCGGTGGAG
TGGGAGAAGAACGGGAAGGCAGAGGACAACACTACAAGACCACGCCG
GCCGTGCTGGACAGCGACGGCTCCTACTTCCTCTACAGCAAGCTC
TCAGTGCCACGAGTGAGTGGCAGCGGGGCGACGTCTTCACCTGC
TCCGTGATGCACGAGGCCTTGCACAACCACTACACGCAGAAGTCC
ATCTCCCGCTCTCCGGGTAAATGA 66
ATGGAGACTGGGCTGCGCTGGCTTCTCCTGGTTCGCTGTGCTCAA
GGTGTCCAGTGTGAGTCGGTGGAGGAGTCCGGAGGAGGCCTGGTA
ACGCCTGGAGGAACCCTGACACTCACCTGCACCGTCTCTGGATT
TCCCTCAGTAACATGCAATGAGCTGGGTCCGCCAGGCTCCAGGG
AAGGGGCTGGAATGGATCGGATTCATTAGTCGTAGTGGTATTACA
TACTACGCGAGCTGGGCAAAAGGCCGATTCACCATCTCCAAAACC
TCGACCACGGTGGATCTGAAAATCACCAAGTCCGACGACCGAGGAC
ACGGCCGCCTATTTCTGTGCCAGAGAATTCGGTGCTGTTGGTAGT
GATTATTATAGGGACGCCTTGAGGTTGTGGGGCCCAGGCACCCTG
GTCACCGTCTCCTCAGGGCAACCTAAGGCTCCATCAGTCTTCCCA
CTGGCCCCCTGCTGCGGGGACACACCCAGCTCCACGGTGACCCTG
GGCTGCCTGGTCAAAGGCTACCTCCCGGAGCCAGTGACCGTGACC
TGGAACCTCGGGCACCCCTACCAATGGGGTACGCACCTTCCCGTCC
GTCCGGCAGTCCTCAGGCCTCTACTCGCTGAGCAGCGTGGTGAGC
GTGACCTCAAGCAGCCAGCCCGTCACCTGCAACGTGGCCCACCCA
GCCACCAACACCAAAGTGGACAAGACCGTTGCGCCCTCGACATGC
AGCAAGCCCACGTGCCCACCCCTGAACTCCTGGGGCGATCCTCT
GTCTTCATCTTCCCCCAAAACCCAAGGACACCCTCATGATCTCA
CGCACCCCGAGGTACATGCGTGGTGGTGGACGTGAGCCAGGAT
GACCCCGAGGTGCAGTTCACATGGTACATAAACACGAGCAGGTG
CGCACCGCCCGGCCGCGCTACGGGAGCAGCAGTTCAACAGCACG
ATCCGCGTGGTCAGCACCCCTCCCCATCGCGCACCAAGGACTGGCTG
AGGGGCAAGGAGTTCAAGTGCAAAGTCCACAACAAGGCACTCCCG
GCCCCCATCGAGAAAACCATCTCCAAAGCCAGAGGGCAGCCCCTG
GAGCCGAAGGTCTACACCATGGGCCCTCCCCGGGAGGAGCTGAGC
AGCAGGTCGGTCAGCCTGACCTGCATGATCAACGGCTTCTACCCT
TCCGACATCTTCGGTGGAGTGGGAGAAGAACGGGAAGGCAGAGGAC
AACTACAAGACCACGCCGGCCGTGCTGGACAGCGACGGCTCCTAC
TTCCTCTACAGCAAGCTCTCAGTGCCACGAGTGAGTGGCAGCGG
GGCGACGTCTTCACCTGCTCCGTGATGCACGAGGCCTTGCACAAC
CACTACACGCAGAAGTCCATCTCCCGCTCTCCGGGTAAATGA 67
ATGGAGACTGGGCTGCGCTGGCTTCTCCTGGTTCGCTGTGCTCAA
GGTGTCCAGTGTGAGTCGGTGGAGGAGTCCGGGGGTCGCCTGGTA
ACGCCTGGGACACCCCTGACACTCACCTGCACAGTCTCTGGAATC
GACCTCAGTACCCATGCAATGACCTGGGTCCGCCAGGCTCCAGGA

AAAGGGGCTGGGATGCGAATGCGGATTAATCCTAGTGGTAGCGCA
TACTACGCGACCTGGGTGAATGGCCGATTACCATCTCCAAAACC
TCGACCACGGTGGATCTGAAAATCACCAGTCCGACAACCGGGGAC
ACGGCCAAGTATTTCTGTGCCAGAGATTATATTACTGCGGGTGAT
TATTATATGGATGCTTTTGTATCCCTGGGGCCCAGGCACCCTGGTC
ACCGTCTCCTCAGGGCAACCTAAGGCTCCATCAGTCTTCCCCTG
GCCCCCTGCTGCGGGGACACACCCAGCTCCACGGTGACCCTGGGC
TGCCTGGTCAAAGGCTACCTCCCGGAGCCAGTGACCGTGACCTGG
AACTCGGGCACCTCACCAATGGGGTACGCACCTTCCCGTCCGTC
CGGCAGTCCTCAGGCCTCTACTCGCTGAGCAGCGTGGTGAGCGTG
ACCTCAAGCAGCCAGCCCGTCACCTGCAACGTGGCCCCACCCAGCC
ACCAACACCAAAGTGGACAAGACCGTTGCGCCCTCGACATGCAGC
AAGCCACAGTGCCCCACCCCTGAACTCCTGGGGCGATCCTCTGTC
TTCATCTTCCCCCCTAAAACCCAAGGACACCCTCATGATCTCACGC
ACCCCCGAGGTCACATGCGTGGTGGTGGACGTGAGCCAGGATGAC
CCCGAGGTGCAGTTCACATGGTACATAAACAACGAGCAGGTGCGC
ACCGCCCCGGCCGCGCTACGGGAGCAGCAGTTCAACAGCACGATC
CGCGTGGTCAGCACCTCCCCATCGCGCACCAAGGACTGGCTGAGG
GGCAAGGAGTTCAAGTGCAAAGTCCACAACAAGGCACTCCCGGCC
CCCATCGAGAAAACCATCTCCAAAGCCAGAGGGCAGCCCCTGGAG
CCGAAGGTCTACACCATGGGGCCCTCCCCGGGAGGAGCTGAGCAGC
AGGTCGGTCAGCCTGACCTGCATGATCAACGGCTTCTACCCTTCC
GACATCTCGGTGGAGTGGGAGAAGAACGGGAAGGCAGAGGACAAC
TACAAGACCACGCCGGCCGTGCTGGACAGCGACGGCTCCTACTTC
CTCTACAGCAAGCTCTCAGTGCCACGAGTGAGTGGCAGCGGGGC
GACGTCTTCACCTGCTCCGTGATGCACGAGGCCTTGCACAACCAC
TACACGCAGAAGTCCATCTCCCGCTCTCCGGGTAAATGA

TABLE-US-00009 TABLE 9 Nucleic acid sequences encoding light chains SEQ ID
NO: Nucleic acid sequences encoding light chains 68

ATGGACACGAGGGCCCCCACTCAGCTGCTGGGGCTCCTGCTGCTC
TGGCTCCCAGGTGCCACATTTGCCCAAGTGCTGACCCAGACTGCA
TCCCCCGTGTCTGCGGCTGTTGGAGGCACAGTCACCATCAATTGC
CAGTCCAGTCAGAGTGTTGTATATAACAACCGCTTATCCTGGTTT
CAACAGAAACCAGGGCAGCCTCCCAAGCTCCTGATCTATGGTGCA
TCCACTCTGGCATCTGGGGTCCCATCGCGGTTCAAAGGCAGTGGA
TCTGGGACACAGTTCACCTCTACCATCAGCGACGTGCAGTGTGAC
GATGCTGCCACTTACTACTGTCTAGGCTCCTATGATTGTAGTAGT
GGTGATTGCCATGCTTTTCGGCGGAGGGACCGAGGTGGTGGTCAAA
GGTGATCCAGTTGCACCTACTGTCCTCATCTTCCCACCAGCTGCT
GATCAGGTGGCAACTGGAACAGTCACCATCGTGTGTGTGGCGAAT
AAATACTTTCCCGATGTCACCGTCACCTGGGAGGTGGATGGCACC
ACCCAAACAACCTGGCATCGAGAACAGTAAAACACCGCAGAATTCT
GCAGATTGTACCTACAACCTCAGCAGCACTCTGACACTGACCAGC
ACACAGTACAACAGCCACAAAGAGTACACCTGCAAGGTGACCCAG
GGCACGACCTCAGTCGTCCAGAGCTTCAATAGGGGTGACTGTTAG 69
ATGGACATGAGGGCCCCCACTCAGCTGCTGGGGCTCCTACTGCTC
TGGCTCCCAGGTGCCAGATGTGCTGACATTGTGATGACCCAGACT
CCAGCCTCCGTGGAGGCAGCTGTGGGAGGCACAGTCACCATCAAT
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CAGAAACCAGGGCAGCCTCCCAACCTCCTGATCTACAGGGCATCC

GCTGCCACTTCAAGAGTATTATGAGGGTAGTGATATT
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TACAACAGCCACAAAGAGTACACCTGCAAGGTGACCCAGGGGCACG
ACCTCAGTCGTCCAGAGCTTCAATAGGGGTGACTGTTAG 73
ATGGACACGAGGGCCCCCACTCAGCTGCTGGGGCTCCTGCTGCTC
TGGCTCCCAGGTGCCAGATGTGCTGACATCGTGATGACCCAGACT
CCAGCCTCTGTGTCTGCAGCTGTGGGAGGCACAGTCACCATCAAG
TGCCAGGCCAGTCAGAGCATTAGCAACCAACTATCCTGGTATCAG
CAGAAATCAGGGCAGCCTCCCAAGCTCCTGATCTACAGGGCATCT
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GGGACAGAGTTCCTCTCACCATCAGCGACCTGGAGTGTGCCGAT
GCTGCCACTTACTACTGTCAACAGGGTTATAATAGAGATAATGTT
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TTTCCCGATGTCACCGTCACCTGGGAGGTGGATGGCACCACCCAA
ACAACCTGGCATCGAGAACAGTAAAACACCGCAGAATTCTGCAGAT
TGTACCTACAACCTCAGCAGCACTCTGACACTGACCAGCACACAG
TACAACAGCCACAAAGAGTACACCTGCAAGGTGACCCAGGGGCACG
ACCTCAGTCGTCCAGAGCTTCAATAGGGGTGACTGTTAG

Methods of the Disclosure

[0086] Disclosed herein are methods for detecting phosphorylated tau in a sample from an individual using antibodies described herein. In some embodiments, the phosphorylated tau is selected from the group consisting of pTau-212, pTau-217, pTau-231, pTau-214, and pTau-220. In some embodiments, methods for detecting phosphorylated tau in a sample from an individual using antibodies described herein comprise improved specificity and sensitivity.

[0087] Described herein are methods for detecting phosphorylated tau in a sample from an individual comprising: performing an assay on the sample using an antibody or antibody fragment that binds to phosphorylated tau. Described herein are methods for detecting phosphorylated tau in a sample from an individual comprising: performing an immunoassay on the sample using an antibody or antibody fragment that binds to phosphorylated tau. In some embodiments, the phosphorylated tau is selected from the group consisting of pTau-212, pTau-217, pTau-231, pTau-214, pTau-220, and pTau-181. In some embodiments, the phosphorylated tau is selected from the group consisting of pTau-212, pTau-217, pTau-231, pTau-214, and pTau-220. In some embodiments, the phosphorylated tau is pTau-217. In some embodiments, the phosphorylated tau is pTau-231. In some embodiments, the phosphorylated tau is pTau-181. In some embodiments, the phosphorylated tau is pTau-212. In some embodiments, the phosphorylated tau is pTau-217. In some embodiments, the phosphorylated tau is pTau-214. In some embodiments, the phosphorylated tau is pTau-220. In some embodiments, the phosphorylated tau is pTau-181 and pTau-217. In some embodiments, the phosphorylated tau is pTau-181 and pTau-231. In some embodiments, the phosphorylated tau is pTau-217 and pTau-231. In some embodiments, the phosphorylated tau is pTau-181, pTau-217, and pTau-231.

[0088] Further described herein are methods for detecting phosphorylated tau in a sample from an individual comprising: performing an assay on the sample using an antibody or antibody fragment that binds to multiple phosphorylated tau proteins. In some embodiments, the methods detects

pTau-217 and pTau-231. In some embodiments, the methods detects pTau-212 and pTau-217. In some embodiments, the methods detects pTau-212 and pTau-231. In some embodiments, the methods detects pTau-212, pTau-217 and pTau-231.

[0089] Described herein are methods for detecting phosphorylated tau in a sample from an individual, wherein the method detects pTau-217 and pTau-231 in a sample selected from the group consisting of a plasma sample and serum sample. In some embodiments, the methods detect pTau-212 and pTau-217 in a sample selected from the group consisting of a plasma sample and serum sample. In some embodiments, the methods detect pTau-212 and pTau-231 in a sample selected from the group consisting of a plasma sample and serum sample. In some embodiments, the methods detect pTau-212, pTau-217, and pTau-231 in a sample selected from the group consisting of a plasma sample and serum sample.

[0090] Methods as described herein can comprise performing an assay on a sample, wherein the sample is selected from the group consisting of a plasma sample and serum sample. In some instances, the sample is a blood sample. In some instances, the sample is a cerebrospinal fluid sample. The sample can be a blood sample obtained by a venous blood draw. The sample can be a blood sample obtained from a finger prick blood draw. The sample can be obtained by a health care provider or by the subject. The method can comprise obtaining a sample from a subject. In some cases, the sample is obtained from the subject during a visit to the clinic or the hospital.

[0091] Further described herein, in some embodiments, are methods to determine a level of a biomarker selected from the group consisting of A042, A040, A038, BACE1, hFABP, TREM2, YKL-40, IP-10, neurogranin, SNAP-25, synaptotagmin, alpha-synuclein, TDP-43, ferritin, VILIP-1, NfL, GFAP, and combinations thereof. In some instances, the biomarker is A042. In some instances, the biomarker is A040. In some instances, the biomarker is A β .sub.42 and A040. In some instances, the biomarker is APOE. In some instances, the biomarker is selected from the group consisting of APOE2, APOE3, and APOE4. In some instances, the biomarker is APOE4.

[0092] In some embodiments, methods for detecting phosphorylated tau in a sample comprise an immunoassay or a ligand assay using the antibodies or antibody fragments described herein. In some cases, the assay is selected from the group consisting of enzyme-linked immunosorbent assay (ELISA), a colorimetric immunoassay, a homogeneous immunoassay, a non-optical immunoassay, a fluorescence immunoassay, a chemiluminescence immunoassay, an electro-chemiluminescence immunoassay, a fluorescence resonance energy transfer (FRET) immunoassay, a time resolved fluorescence immunoassay, a lateral flow immunoassay, a microspot immunoassay, a surface plasmon resonance assay, a ligand assay, a clotting assay, a chromatography assay, and immunocapture coupled with mass spectrometry. In some cases, the assay comprises an immunoassay. In some cases, the assay is selected from the group consisting of a Western blot, enzyme-linked immunosorbent assays (ELISA), and chromatography. In some cases, the immunoassays are single-plexed. In some cases, the immunoassays are multiplexed.

[0093] Methods as described herein can comprise a plurality of immunoassays using the antibodies or antibody fragments described herein. In some cases, the plurality of immunoassays are the same immunoassay (e.g., four or more ELISA assays). When the plurality of immunoassays are the same immunoassay, each of the plurality of immunoassays can detect a different phosphorylated tau.

When the plurality of immunoassays are the same immunoassay, each of the plurality of immunoassays can be performed in the same reaction chamber or a different reaction chamber. A reaction chamber can be any suitable space for performing an immunoassay. Examples of reaction chambers include, but are not limited to, a well in a microplate, an Eppendorf tube, or a droplet.

[0094] In some cases, the plurality of immunoassays are different immunoassays. When the plurality of immunoassays are different immunoassays, each of the plurality of immunoassays can detect a different phosphorylated tau. When the plurality of immunoassays are different immunoassays, each of the plurality of immunoassays can be performed in the same reaction chamber or a different reaction chamber.

[0095] In some cases, the assay comprises a non-immunoassay. In some cases, the assay is selected from the group consisting of High Performance Liquid Chromatography (HPLC), High Performance Liquid Chromatography Mass spectrometry (HPLC-MS), Gas Chromatography Mass Spectrometry (GC-MS), Liquid Chromatography Mass spectrometry (LC-MS), Liquid Chromatography Tandem Mass spectrometry (LC-MS/MS), immunohistochemistry (IHC), polymerase chain reaction (PCR), quantitative PCR (qPCR), and combinations thereof.

[0096] Methods as described herein using the antibodies described herein may be used for establishing Alzheimer's disease in the individual based on detection of phosphorylated tau. In some embodiments, Alzheimer's disease in the individual is established if pTau-212, pTau-217, pTau-231, pTau-214, pTau-220, or combinations thereof is detected in the sample from the individual.

[0097] Methods as described herein using the antibodies described herein may be used for prognosis of the individual for developing Alzheimer's disease based on detection of phosphorylated tau. In some embodiments, prognosis of the individual for developing Alzheimer's disease is determined if pTau-212, pTau-217, pTau-231, pTau-214, pTau-220, or combinations thereof is detected in the sample from the individual.

[0098] Methods as described herein using the antibodies described herein may be used accurately and specifically establish Alzheimer's disease (AD) in an individual as compared to a disease or disorder or neurologically and cognitively unimpaired condition, selected from the group consisting of a non-Alzheimer's disease(AD) neurodegenerative disease, a AD-negative non-AD neurodegenerative disease, a AD-positive non-AD neurodegenerative diseases, behavioral variant of frontotemporal dementia (BvFTD), primary progressive aphasia (PPA), vascular dementia (VaD), Parkinson's disease (PD), PD with dementia (PDD), multiple system atrophy (MSA), progressive supranuclear palsy (PSP), corticobasal syndrome (CBS), AR-negative cognitively impaired or unimpaired controls and combinations thereof. In some embodiments, the methods as described herein using the antibodies described herein comprise an improved accuracy or specificity of at least or about 70%, 80%, 90%, 95%, 99%, or more at establishing AD as compared to a disease or disorder or neurologically and cognitively unimpaired condition.

[0099] Methods as described herein using the antibodies described herein may be used accurately and specifically establish Alzheimer's disease (AD) in an individual as compared to a neuropathological examination or clinical diagnosis. In some embodiments, the methods as described herein using the antibodies described herein comprise an improved accuracy or specificity of at least or about 70%, 80%, 90%, 95%, 99%, or more at establishing AD as compared to a neuropathological examination or clinical diagnosis. In some embodiments, the neuropathological examination or clinical diagnosis comprises neurological tests, mental exams, or brain imaging (e.g. MRI, CT, or PET scans).

[0100] Methods as described herein using the antibodies described herein may be capable of detecting phosphorylated tau in the sample at a low limit of detection. In some embodiments, the methods as described herein using the antibodies described herein are capable of detecting phosphorylated tau in the sample at a limit of detection of at least about 1.5 picogram per milliliter (pg/mL). In some embodiments, the methods as described herein using the antibodies described herein are capable of detecting phosphorylated tau in the sample at a limit of detection of at least about 5 picogram per milliliter (pg/mL). In some embodiments, the methods as described herein using the antibodies described herein are capable of detecting phosphorylated tau in the sample at a limit of detection in a range of about 0.5 pg/mL to about 10 µg/mL, about 1 µg/mL to about 8 µg/mL, about 1.5 pg/mL to about 7 µg/mL, about 2 µg/mL to about 6 µg/mL, or about 3 µg/mL to about 5 µg/mL.

Production of Tau Antibodies

[0101] In some embodiments, antibodies or antibody fragments described herein are produced using any method known in the art to be useful for the synthesis of antibodies or antibody

fragments, in particular, by chemical synthesis or by recombinant expression, and are preferably produced by recombinant expression techniques.

[0102] In some instances, an antibody or its binding fragment thereof is expressed recombinantly, and the nucleic acid encoding the antibody or its binding fragment is assembled from chemically synthesized oligonucleotides (e.g., as described in Kutmeier et al., 1994, *BioTechniques* 17:242), which involves the synthesis of overlapping oligonucleotides containing portions of the sequence encoding the antibody, annealing and ligation of those oligonucleotides, and then amplification of the ligated oligonucleotides by PCR.

[0103] Alternatively, a nucleic acid molecule encoding an antibody is optionally generated from a suitable source (e.g., an antibody cDNA library, or cDNA library generated from any tissue or cells expressing the immunoglobulin) by PCR amplification using synthetic primers hybridizable to the 3' and 5' ends of the sequence or by cloning using an oligonucleotide probe specific for the particular gene sequence.

[0104] In some instances, an antibody or its binding is optionally generated by immunizing an animal, such as a mouse, to generate polyclonal antibodies or, more preferably, by generating monoclonal antibodies, e.g., as described by Kohler and Milstein (1975, *Nature* 256:495-497) or, as described by Kozbor et al. (1983, *Immunology Today* 4:72) or Cole et al. (1985 in *Monoclonal Antibodies and Cancer Therapy*, Alan R. Liss, Inc., pp. 77-96). Alternatively, a clone encoding at least the Fab portion of the antibody is optionally obtained by screening Fab expression libraries (e.g., as described in Huse et al., 1989, *Science* 246:1275-1281) for clones of Fab fragments that bind the specific antigen or by screening antibody libraries (See, e.g., Clackson et al., 1991, *Nature* 352:624; Hane et al., 1997 *Proc. Natl. Acad. Sci. USA* 94:4937).

[0105] In some embodiments, techniques developed for the production of "chimeric antibodies" (Morrison et al., 1984, *Proc. Natl. Acad. Sci.* 81:851-855; Neuberger et al., 1984, *Nature* 312:604-608; Takeda et al., 1985, *Nature* 314:452-454) by splicing genes from a mouse antibody molecule of appropriate antigen specificity together with genes from a human antibody molecule of appropriate biological activity are used. A chimeric antibody is a molecule in which different portions are derived from different animal species, such as those having a variable region derived from a murine monoclonal antibody and a human immunoglobulin constant region.

[0106] In some embodiments, techniques described for the production of single chain antibodies (U.S. Pat. No. 4,694,778; Bird, 1988, *Science* 242:423-42; Huston et al., 1988, *Proc. Natl. Acad. Sci. USA* 85:5879-5883; and Ward et al., 1989, *Nature* 334:544-54) are adapted to produce single chain antibodies. Single chain antibodies are formed by linking the heavy and light chain fragments of the Fv region via an amino acid bridge, resulting in a single chain polypeptide. Techniques for the assembly of functional Fv fragments in *E. coli* are also optionally used (Skerra et al., 1988, *Science* 242:1038-1041).

[0107] In some embodiments, an expression vector comprising the nucleotide sequence of an antibody or the nucleotide sequence of an antibody is transferred to a host cell by conventional techniques (e.g., electroporation, liposomal transfection, and calcium phosphate precipitation), and the transfected cells are then cultured by conventional techniques to produce the antibody. In specific embodiments, the expression of the antibody is regulated by a constitutive, an inducible or a tissue, specific promoter.

[0108] In some embodiments, a variety of host-expression vector systems is utilized to express an antibody, or its binding fragment described herein. Such host-expression systems represent vehicles by which the coding sequences of the antibody is produced and subsequently purified, but also represent cells that are, when transformed or transfected with the appropriate nucleotide coding sequences, express an antibody or its binding fragment in situ. These include, but are not limited to, microorganisms such as bacteria (e.g., *E. coli* and *B. subtilis*) transformed with recombinant bacteriophage DNA, plasmid DNA or cosmid DNA expression vectors containing an antibody or its binding fragment coding sequences; yeast (e.g., *Saccharomyces Pichia*) transformed with

recombinant yeast expression vectors containing an antibody or its binding fragment coding sequences; insect cell systems infected with recombinant virus expression vectors (e.g., baculovirus) containing an antibody or its binding fragment coding sequences; plant cell systems infected with recombinant virus expression vectors (e.g., cauliflower mosaic virus (CaMV) and tobacco mosaic virus (TMV)) or transformed with recombinant plasmid expression vectors (e.g., Ti plasmid) containing an antibody or its binding fragment coding sequences; or mammalian cell systems (e.g., COS, CHO, BH, 293, 293T, 3T3 cells) harboring recombinant expression constructs containing promoters derived from the genome of mammalian cells (e.g., metallothionein promoter) or from mammalian viruses (e.g. the adenovirus late promoter; the vaccinia virus 7.5K promoter).

[0109] For long-term, high-yield production of recombinant proteins, stable expression is preferred. In some instances, cell lines that stably express an antibody are optionally engineered. Rather than using expression vectors that contain viral origins of replication, host cells are transformed with DNA controlled by appropriate expression control elements (e.g., promoter, enhancer, sequences, transcription terminators, polyadenylation sites, etc.), and a selectable marker. Following the introduction of the foreign DNA, engineered cells are then allowed to grow for 1-2 days in an enriched media, and then are switched to a selective media. The selectable marker in the recombinant plasmid confers resistance to the selection and allows cells to stably integrate the plasmid into their chromosomes and grow to form foci that in turn are cloned and expanded into cell lines. This method can advantageously be used to engineer cell lines which express the antibody or its binding fragments.

[0110] In some instances, a number of selection systems are used, including but not limited to the herpes simplex virus thymidine kinase (Wigler et al., 1977, Cell 11:223), hypoxanthine-guanine phosphoribosyltransferase (Szybalska & Szybalski, 1972, Proc. Natl. Acad. Sci. USA 48:202), and adenine phosphoribosyltransferase (Lowy et al., 1980, Cell 22:817) genes are employed in tk-, hgp^rt- or ap^rt-cells, respectively. Also, antimetabolite resistance are used as the basis of selection for the following genes: dhfr, which confers resistance to methotrexate (Wigler et al., 1980, Proc. Natl. Acad. Sci. USA 77:357; O'Hare et al., 1981, Proc. Natl. Acad. Sci. USA 78:1527); gpt, which confers resistance to mycophenolic acid (Mulligan & Berg, 1981, Proc. Natl. Acad. Sci. USA 78:2072); neo, which confers resistance to the aminoglycoside G-418 (Clinical Pharmacy 12:488-505; Wu and Wu, 1991, Biotherapy 3:87-95; Tolstoshev, 1993, Ann. Rev. Pharmacol. Toxicol. 32:573-596; Mulligan, 1993, Science 260:926-932; and Morgan and Anderson, 1993, Ann. Rev. Biochem. 62:191-217; May 1993, TIB TECH 11(5):155-215) and hyg^r, which confers resistance to hygromycin (Santerre et al., 1984, Gene 30:147). Methods commonly known in the art of recombinant DNA technology which can be used are described in Ausubel et al. (eds., 1993, Current Protocols in Molecular Biology, John Wiley & Sons, NY; Kriegler, 1990, Gene Transfer and Expression, A Laboratory Manual, Stockton Press, NY; and in Chapters 12 and 13, Dracopoli et al. (eds), 1994, Current Protocols in Human Genetics, John Wiley & Sons, NY.; Colberre-Garapin et al., 1981, J. Mol. Biol. 150:1).

[0111] In some instances, the expression levels of an antibody are increased by vector amplification (for a review, see Bebbington and Hentschel, the use of vectors based on gene amplification for the expression of cloned genes in mammalian cells in DNA cloning, Vol. 3. (Academic Press, New York, 1987)). When a marker in the vector system expressing an antibody is amplifiable, an increase in the level of inhibitor present in culture of host cell will increase the number of copies of the marker gene. Since the amplified region is associated with the nucleotide sequence of the antibody, production of the antibody will also increase (Crouse et al., 1983, Mol. Cell Biol. 3:257).

[0112] In some instances, any method known in the art for purification of an antibody is used, for example, by chromatography (e.g., ion exchange, affinity, particularly by affinity for the specific antigen after Protein A, and sizing column chromatography), centrifugation, differential solubility, or by any other standard technique for the purification of proteins.

Expression Vectors

[0113] In some embodiments, vectors include any suitable vectors derived from either a eukaryotic or prokaryotic sources. In some cases, vectors are obtained from bacteria (e.g. *E. coli*), insects, yeast (e.g. *Pichia pastoris*), algae, or mammalian sources. Exemplary bacterial vectors include pACYC177, pASK75, pBAD vector series, pBADM vector series, pET vector series, pETM vector series, pGEX vector series, pHAT, pHAT2, pMal-c2, pMal-p2, pQE vector series, pRSET A, pRSET B, pRSET C, pTrcHis2 series, pZA31-Luc, pZE21-MCS-1, pFLAG ATS, pFLAG CTS, pFLAG MAC, pFLAG Shift-12c, pTAC-MAT-1, pFLAG CTC, or pTAC-MAT-2.

[0114] Exemplary insect vectors include pFastBac1, pFastBac DUAL, pFastBac ET, pFastBac HTa, pFastBac HTb, pFastBac HTc, pFastBac M30a, pFastBac M30b, pFastBac M30c, pVL1392, pVL1393, pVL1393 M10, pVL1393 M11, pVL1393 M12, FLAG vectors such as pPolh-FLAG1 or pPolh-MAT 2, or MAT vectors such as pPolh-MAT1, or pPolh-MAT2.

[0115] In some cases, yeast vectors include Gateway® pDESTTM 14 vector, Gateway® pDESTTM 15 vector, Gateway® pDESTTM 17 vector, Gateway® pDESTTM 24 vector, Gateway® pYES-DEST52 vector, pBAD-DEST49 Gateway® destination vector, pAO815 *Pichia* vector, pFLD1 *Pichi pastoris* vector, pGAPZA, B, & C *Pichia pastoris* vector, pPIC3.5K *Pichia* vector, pPIC6 A, B, & C *Pichia* vector, pPIC9K *Pichia* vector, pTEF1/Zeo, pYES2 yeast vector, pYES2/CT yeast vector, pYES2/NT A, B, & C yeast vector, or pYES3/CT yeast vector.

[0116] Exemplary algae vectors include pChlamy-4 vector or MCS vector.

[0117] Examples of mammalian vectors include transient expression vectors or stable expression vectors. Mammalian transient expression vectors may include pRK5, p3×FLAG-CMV 8, pFLAG-Myc-CMV 19, pFLAG-Myc-CMV 23, pFLAG-CMV 2, pFLAG-CMV 6a,b,c, pFLAG-CMV 5.1, pFLAG-CMV 5a,b,c, p3×FLAG-CMV 7.1, pFLAG-CMV 20, p3×FLAG-Myc-CMV 24, pCMV-FLAG-MAT1, pCMV-FLAG-MAT2, pBICEP-CMV 3, or pBICEP-CMV 4. Mammalian stable expression vector may include pFLAG-CMV 3, p3×FLAG-CMV 9, p3×FLAG-CMV 13, pFLAG-Myc-CMV 21, p3×FLAG-Myc-CMV 25, pFLAG-CMV 4, p3×FLAG-CMV 10, p3×FLAG-CMV 14, pFLAG-Myc-CMV 22, p3×FLAG-Myc-CMV 26, pBICEP-CMV 1, or pBICEP-CMV 2.

[0118] In some instances, a cell-free system is a mixture of cytoplasmic and/or nuclear components from a cell and is used for in vitro nucleic acid synthesis. In some cases, a cell-free system utilizes either prokaryotic cell components or eukaryotic cell components. Sometimes, a nucleic acid synthesis is obtained in a cell-free system based on for example *Drosophila* cell, *Xenopus* egg, or HeLa cells. Exemplary cell-free systems include, but are not limited to, *E. coli* S30 Extract system, *E. coli* T7 S30 system, or PURExpress®.

Host Cells

[0119] In some embodiments, a host cell includes any suitable cell such as a naturally derived cell or a genetically modified cell. In some instances, a host cell is a production host cell. In some instances, a host cell is a eukaryotic cell. In other instances, a host cell is a prokaryotic cell. In some cases, a eukaryotic cell includes fungi (e.g., yeast cells), animal cell or plant cell. In some cases, a prokaryotic cell is a bacterial cell. Examples of bacterial cell include gram-positive bacteria or gram-negative bacteria. Sometimes the gram-negative bacteria is anaerobic, rod-shaped, or both.

[0120] In some instances, gram-positive bacteria include Actinobacteria, Firmicutes or Tenericutes. In some cases, gram-negative bacteria include Aquificae, Deinococcus-Thermus, Fibrobacteres-Chlorobi/Bacteroidetes (FCB group), Fusobacteria, Gemmatimonadetes, Nitrospirae, Planctomycetes-Verrucomicrobia/Chlamydiae (PVC group), Proteobacteria, Spirochaetes or Synergistetes. Other bacteria can be Acidobacteria, Chloroflexi, Chrysiogenetes, Cyanobacteria, Deferribacteres, Dictyoglomi, Thermodesulfobacteria or Thermotogae. A bacterial cell can be *Escherichia coli*, *Clostridium botulinum*, or *Coli* bacilli.

[0121] Exemplary prokaryotic host cells include, but are not limited to, BL21, MachiTM, DH10BTM, TOP10, DH5α, DH10BacTM, OmniMaxTM, MegaXTM, DH12STM, INV110, TOP10F', INVαF, TOP10/P3, ccdB Survival, PIR1, PIR2, Stbl2TM, Stbl3TM, or Stbl4TM.

[0122] In some instances, animal cells include a cell from a vertebrate or from an invertebrate. In some cases, an animal cell includes a cell from a marine invertebrate, fish, insects, amphibian, reptile, or mammal. In some cases, a fungus cell includes a yeast cell, such as brewer's yeast, baker's yeast, or wine yeast.

[0123] Fungi include ascomycetes such as yeast, mold, filamentous fungi, basidiomycetes, or zygomycetes. In some instances, yeast includes Ascomycota or Basidiomycota. In some cases, Ascomycota includes Saccharomycotina (true yeasts, e.g. *Saccharomyces cerevisiae* (baker's yeast)) or Taphrinomycotina (e.g. Schizosaccharomycetes (fission yeasts)). In some cases, Basidiomycota includes Agaricomycotina (e.g. Tremellomycetes) or Pucciniomycotina (e.g. Microbotryomycetes).

[0124] Exemplary yeast or filamentous fungi include, for example, the genus: *Saccharomyces*, *Schizosaccharomyces*, *Candida*, *Pichia*, *Hansenula*, *Kluyveromyces*, *Zygosaccharomyces*, *Yarrowia*, *Trichosporon*, *Rhodospiridi*, *Aspergillus*, *Fusarium*, or *Trichoderma*. Exemplary yeast or filamentous fungi include, for example, the species: *Saccharomyces cerevisiae*, *Schizosaccharomyces pombe*, *Candida utilis*, *Candida boidini*, *Candida albicans*, *Candida tropicalis*, *Candida stellatoidea*, *Candida glabrata*, *Candida krusei*, *Candida parapsilosis*, *Candida guilliermondii*, *Candida viswanathii*, *Candida lusitaniae*, *Rhodotorula mucilaginosa*, *Pichia metanolica*, *Pichia angusta*, *Pichia pastoris*, *Pichia anomala*, *Hansenula polymorpha*, *Kluyveromyces lactis*, *Zygosaccharomyces rouxii*, *Yarrowia lipolytica*, *Trichosporon pullulans*, *Rhodospiridium toru-Aspergillus niger*, *Aspergillus nidulans*, *Aspergillus awamori*, *Aspergillus oryzae*, *Trichoderma reesei*, *Yarrowia lipolytica*, *Brettanomyces bruxellensis*, *Candida stellata*, *Schizosaccharomyces pombe*, *Torulaspora delbrueckii*, *Zygosaccharomyces bailii*, *Cryptococcus neoformans*, *Cryptococcus gattii*, or *Saccharomyces boulardii*.

[0125] Exemplary yeast host cells include, but are not limited to, *Pichia pastoris* yeast strains such as GS115, KM71H, SMD1168, SMD1168H, and X-33; and *Saccharomyces cerevisiae* yeast strain such as INVSc1.

[0126] In some instances, additional animal cells include cells obtained from a mollusk, arthropod, annelid or sponge. In some cases, an additional animal cell is a mammalian cell, e.g., from a primate, ape, equine, bovine, porcine, canine, feline or rodent. In some cases, a rodent includes mouse, rat, hamster, gerbil, hamster, chinchilla, fancy rat, or guinea pig.

[0127] Exemplary mammalian host cells include, but are not limited to, 293A cell line, 293FT cell line, 293F cells, 293 H cells, CHO DG44 cells, CHO-S cells, CHO-K1 cells, FUT8 KO CHOK1, Expi293F™ cells, Flp-In™ T-REx™ 293 cell line, Flp-In™-293 cell line, Flp-In™-3T3 cell line, Flp-In™-BHK cell line, Flp-In™-CHO cell line, Flp-In™-CV-1 cell line, Flp-In™-Jurkat cell line, FreeStyle™ 293-F cells, FreeStyle™ CHO-S cells, GripTite™ 293 MSR cell line, GS-CHO cell line, HepaRG™ cells, T-REx™ Jurkat cell line, Per.C6 cells, T-REx™-293 cell line, T-REx™-CHO cell line, and T-REx™-HeLa cell line.

[0128] In some instances, a mammalian host cell is a stable cell line, or a cell line that has incorporated a genetic material of interest into its own genome and has the capability to express the product of the genetic material after many generations of cell division. In some cases, a mammalian host cell is a transient cell line, or a cell line that has not incorporated a genetic material of interest into its own genome and does not have the capability to express the product of the genetic material after many generations of cell division.

[0129] Exemplary insect host cells include, but are not limited to, *Drosophila* S2 cells, Sf9 cells, Sf21 cells, High Five™ cells, and expresSF+® cells.

[0130] In some instances, plant cells include a cell from algae. Exemplary insect cell lines include, but are not limited to, strains from *Chlamydomonas reinhardtii* 137c, or *Synechococcus elongatus* PPC 7942.

NUMBERED EMBODIMENTS

[0131] Numbered embodiment 1 comprises a method for detecting phosphorylated tau in a sample

from an individual comprising: performing an immunoassay on the sample using an antibody or antibody fragment comprising a variable domain, heavy chain region (VH) and a variable domain, light chain region (VL), wherein the VH comprises an amino acid sequence at least about 90% identical to a sequence as set forth in any one of SEQ ID NOs: 30-34, and wherein the VL comprises an amino acid sequence at least about 90% identical to a sequence as set forth in any one of SEQ ID NOs: 35-40. Numbered embodiment 2 comprises the method of numbered embodiment 1, wherein the phosphorylated tau is selected from the group consisting of pTau-181, pTau-212, pTau-217, pTau-231, pTau-214, and pTau-220. Numbered embodiment 3 comprises the method of numbered embodiments 1-2, wherein the phosphorylated tau is pTau-217. Numbered embodiment 4 comprises the method of numbered embodiments 1-2, wherein the phosphorylated tau is pTau-231. Numbered embodiment 5 comprises the method of numbered embodiment 2, wherein the method detects pTau-217 and pTau-231. Numbered embodiment 6 comprises the method of numbered embodiment 2, wherein the method detects pTau-212 and pTau-217. Numbered embodiment 7 comprises the method of numbered embodiment 2, wherein the method detects pTau-212 and pTau-231. Numbered embodiment 8 comprises the method of numbered embodiment 2, wherein the method detects pTau-181 and pTau-217. Numbered embodiment 9 comprises the method of numbered embodiment 2, wherein the method detects pTau-181 and pTau-231. Numbered embodiment 10 comprises the method of numbered embodiment 2, wherein the method detects pTau-181, pTau-217 and pTau-231. Numbered embodiment 11 comprises the method of numbered embodiment 2, wherein the method detects pTau-212, pTau-217 and pTau-231. Numbered embodiment 12 comprises the method of numbered embodiment 5, wherein the method detects pTau-217 and pTau-231 in a sample selected from the group consisting of a plasma sample and serum sample. Numbered embodiment 13 comprises the method of numbered embodiment 6, wherein the method detects pTau-212 and pTau-217 in a sample selected from the group consisting of a plasma sample and serum sample. Numbered embodiment 14 comprises the method of numbered embodiment 7, wherein the method detects pTau-212 and pTau-231 in a sample selected from the group consisting of a plasma sample and serum sample. Numbered embodiment 15 comprises the method of numbered embodiment 11, wherein the method detects pTau-181 and pTau-217 in a sample selected from the group consisting of a plasma sample and serum sample. Numbered embodiment 16 comprises the method of numbered embodiment 11, wherein the method detects pTau-181 and pTau-231 in a sample selected from the group consisting of a plasma sample and serum sample. Numbered embodiment 17 comprises the method of numbered embodiment 11, wherein the method detects pTau-181, pTau-217, and pTau-231 in a sample selected from the group consisting of a plasma sample and serum sample. Numbered embodiment 18 comprises the method of numbered embodiment 11, wherein the method detects pTau-212, pTau-217, and pTau-231 in a sample selected from the group consisting of a plasma sample and serum sample. Numbered embodiment 19 comprises the method of numbered embodiments 1-18, wherein the VH comprises an amino acid sequence according to any one of SEQ ID NOs: 30-34. Numbered embodiment 20 comprises the method of numbered embodiments 1-19, wherein the VL comprises an amino acid sequence according to any one of SEQ ID NOs: 35-40. Numbered embodiment 21 comprises the method of numbered embodiments 1-20, wherein the VH comprises an amino acid sequence according to any one of SEQ ID NOs: 30-34, and wherein the VL comprises an amino acid sequence according to any one of SEQ ID NOs: 35-40. Numbered embodiment 22 comprises the method of numbered embodiments 1-21, wherein the VH comprises an amino acid sequence at least about 90% identical to SEQ ID NO: 30, and wherein the VL comprises an amino acid sequence at least about 90% identical to SEQ ID NO: 35. Numbered embodiment 23 comprises the method of numbered embodiments 1-21, wherein the VH comprises an amino acid sequence at least about 90% identical to SEQ ID NO: 31, and wherein the VL comprises an amino acid sequence at least about 90% identical to SEQ ID NO: 36. Numbered embodiment 24 comprises the method of numbered embodiments 1-21, wherein the VH comprises

an amino acid sequence at least about 90% identical to SEQ ID NO: 31, and wherein the VL comprises an amino acid sequence at least about 90% identical to SEQ ID NO: 37. Numbered embodiment 25 comprises the method of numbered embodiments 1-21, wherein the VH comprises an amino acid sequence at least about 90% identical to SEQ ID NO: 32, and wherein the VL comprises an amino acid sequence at least about 90% identical to SEQ ID NO: 38. Numbered embodiment 26 comprises the method of numbered embodiments 1-21, wherein the VH comprises an amino acid sequence at least about 90% identical to SEQ ID NO: 33, and wherein the VL comprises an amino acid sequence at least about 90% identical to SEQ ID NO: 39. Numbered embodiment 27 comprises the method of numbered embodiments 1-21, wherein the VH comprises an amino acid sequence at least about 90% identical to SEQ ID NO: 34, and wherein the VL comprises an amino acid sequence at least about 90% identical to SEQ ID NO: 40. Numbered embodiment 28 comprises the method of numbered embodiments 1-27, wherein the antibody or antibody fragment comprises an amino acid sequence at least about 90% identical to any one of SEQ ID NOs: 41-51. Numbered embodiment 29 comprises the method of numbered embodiments 1-28, further comprising performing an assay on the sample to determine a level of a biomarker selected from the group consisting of A042, A040, A038, BACE1, hFABP, TREM2, YKL-40, IP-10, neurogranin, SNAP-25, synaptotagmin, alpha-synuclein, TDP-43, ferritin, VILIP-1, NfL, GFAP, and combinations thereof. Numbered embodiment 30 comprises the method of numbered embodiments 1-29, wherein the sample is selected from the group consisting of a blood sample, a plasma sample, a serum sample, and a cerebrospinal fluid (CSF) sample. Numbered embodiment 31 comprises the method of numbered embodiments 1-30, further comprising establishing Alzheimer's disease in the individual based on detection of phosphorylated tau. Numbered embodiment 32 comprises the method of numbered embodiments 1-31, further comprising establishing prognosis of the individual for developing Alzheimer's disease based on detection of phosphorylated tau. Numbered embodiment 33 comprises the method of numbered embodiment 32, further determining the individual's age, genotype, or expression of a biomarker. Numbered embodiment 34 comprises the method of numbered embodiment 33, wherein the biomarker is selected from the group consisting of A042, A040, A038, BACE1, hFABP, TREM2, YKL-40, IP-10, neurogranin, SNAP-25, synaptotagmin, alpha-synuclein, TDP-43, ferritin, VILIP-1, NfL, GFAP, and combinations thereof. Numbered embodiment 35 comprises the method of numbered embodiments 1-34, wherein the method has a specificity of at least about 80% for detecting phosphorylated tau. Numbered embodiment 36 comprises the method of numbered embodiments 1-34, wherein the method has a specificity of at least about 85% for detecting phosphorylated tau. Numbered embodiment 37 comprises the method of numbered embodiments 1-34, wherein the method has a specificity of at least about 90% for detecting phosphorylated tau. Numbered embodiment 38 comprises the method of numbered embodiments 1-37, wherein the method has a sensitivity of at least about 80% for detecting phosphorylated tau. Numbered embodiment 39 comprises the method of numbered embodiments 1-37, wherein the method has a sensitivity of at least about 85% for detecting phosphorylated tau. Numbered embodiment 40 comprises the method of numbered embodiments 1-37, wherein the method has a sensitivity of at least about 90% for detecting phosphorylated tau. Numbered embodiment 41 comprises the method of numbered embodiments 1-40, wherein the method is capable of detecting phosphorylated tau in the sample at a limit of detection of at least about 1.0 picogram per milliliter (pg/mL). Numbered embodiment 42 comprises the method of numbered embodiments 1-40, wherein the method is capable of detecting phosphorylated tau in the sample at a limit of detection of at least about 1.5 picogram per milliliter (pg/mL). Numbered embodiment 43 comprises the method of numbered embodiments 1-40, wherein the method is capable of detecting phosphorylated tau in the sample at a limit of detection of at least about 5 picogram per milliliter (pg/mL).

[0132] Numbered embodiment 44 comprises an anti-tau antibody comprising i) a heavy chain comprising variable heavy chain (VH) domain and ii) a light chain comprising a variable light

chain (VL) domain, wherein the VH domain comprises HCDR1 sequence comprising a sequence selected from SEQ ID NOs: 1-5, HCDR2 sequence comprising a sequence selected from SEQ ID NOs: 6-9, and HCDR3 sequence comprising a sequence selected from SEQ ID NOs: 10-13, and VL domain comprises LCDR1 sequence comprising a sequence selected from SEQ ID NOs: 14-19, LCDR2 sequence comprising a sequence selected from SEQ ID NOs: 20-23, and LCDR3 sequence comprising a sequence selected from SEQ ID NOs: 24-29. Numbered embodiment 45 comprises the anti-tau antibody of numbered embodiment 44, wherein the HCDR1 sequence comprises SEQ ID NO: 1, HCDR2 sequence comprises SEQ ID NO: 6, HCDR3 sequence comprises SEQ ID NO: 10, LCDR1 sequence comprises SEQ ID NO: 14, LCDR2 sequence comprises SEQ ID NO: 20, and LCDR3 sequence comprises SEQ ID NO: 24. Numbered embodiment 46 comprises the anti-tau antibody of numbered embodiment 44, wherein the HCDR1 sequence comprises SEQ ID NO: 2, HCDR2 sequence comprises SEQ ID NO: 7, HCDR3 sequence comprises SEQ ID NO: 11, LCDR1 sequence comprises SEQ ID NO: 15, LCDR2 sequence comprises SEQ ID NO: 21, and LCDR3 sequence comprises SEQ ID NO: 25. Numbered embodiment 47 comprises the anti-tau antibody of numbered embodiment 44, wherein the HCDR1 sequence comprises SEQ ID NO: 2, HCDR2 sequence comprises SEQ ID NO: 7, HCDR3 sequence comprises SEQ ID NO: 11, LCDR1 sequence comprises SEQ ID NO: 16, LCDR2 sequence comprises SEQ ID NO: 22, and LCDR3 sequence comprises SEQ ID NO: 26. Numbered embodiment 48 comprises the anti-tau antibody of numbered embodiment 44, wherein the HCDR1 sequence comprises SEQ ID NO: 3, HCDR2 sequence comprises SEQ ID NO: 8, HCDR3 sequence comprises SEQ ID NO: 10, LCDR1 sequence comprises SEQ ID NO: 17, LCDR2 sequence comprises SEQ ID NO: 20, and LCDR3 sequence comprises SEQ ID NO: 27. Numbered embodiment 49 comprises the anti-tau antibody of numbered embodiment 44, wherein the HCDR1 sequence comprises SEQ ID NO: 4, HCDR2 sequence comprises SEQ ID NO: 7, HCDR3 sequence comprises SEQ ID NO: 12, LCDR1 sequence comprises SEQ ID NO: 18, LCDR2 sequence comprises SEQ ID NO: 23, and LCDR3 sequence comprises SEQ ID NO: 28. Numbered embodiment 50 comprises the anti-tau antibody of numbered embodiment 44, wherein the HCDR1 sequence comprises SEQ ID NO: 5, HCDR2 sequence comprises SEQ ID NO: 9, HCDR3 sequence comprises SEQ ID NO: 13, LCDR1 sequence comprises SEQ ID NO: 19, LCDR2 sequence comprises SEQ ID NO: 21, and LCDR3 sequence comprises SEQ ID NO: 29. Numbered embodiment 51 comprises the anti-tau antibody of numbered embodiment 44, wherein the VH domain comprises at least 80%, at least 85%, at least 90%, at least 95% sequence identity to a sequence selected from SEQ ID NOs: 30-34. Numbered embodiment 52 comprises the anti-tau antibody of numbered embodiment 44, wherein the VL domain comprises at least 80%, at least 85%, at least 90%, at least 95% sequence identity to a sequence selected from SEQ ID NOs: 35-40.

[0133] Numbered embodiment 53 comprises the anti-tau antibody of numbered embodiments 44-52, wherein the anti-tau antibody is a chimeric antibody or antigen binding fragment thereof. Numbered embodiment 54 comprises the anti-tau antibody of numbered embodiments 44-53, wherein the anti-tau antibody comprises an IgG-scFv, nanobody, BiTE, diabody, DART, TandAb, scDiabody, scDiabody-CH3, triple body, mini-antibody, minibody, TriBi minibody, scFv-CH3 KIH, Fab-scFv-Fc KIH, Fab-scFv, scFv-CH-CL-scFv, Fab', F(ab')₂, F(ab')₃, F(ab')₂-scFv₂, scFv, scFv-KIH, Fab-scFv-Fc, tetravalent HCAb, scDiabody-Fc, diabody-Fc, tandem scFv-Fc, or intrabody. Numbered embodiment 55 comprises the anti-tau antibody of numbered embodiments 44-54, wherein the anti-tau antibody is an IgG1 antibody. Numbered embodiment 56 comprises the anti-tau antibody of numbered embodiments 44-55, wherein the anti-tau antibody is an IgG2 antibody. Numbered embodiment 57 comprises the anti-tau antibody of numbered embodiments 44-56, wherein the anti-tau antibody is an IgG4 antibody. Numbered embodiment 58 comprises the anti-tau antibody of numbered embodiments 44-57, wherein the light chain is a kappa chain. Numbered embodiment 59 comprises the anti-tau antibody of numbered embodiments 44-58, wherein the anti-tau antibody has a binding affinity to human tau of about 100 pM to about 3 nM. Numbered

embodiment 60 comprises the anti-tau antibody of numbered embodiments 44-59, wherein the anti-tau antibody comprises a VH domain that is encoded by a nucleic acid comprising at least 80%, at least 85%, at least 90%, at least 95% sequence identity to a sequence selected from SEQ ID NOs: 52-56. Numbered embodiment 61 comprises the anti-tau antibody of numbered embodiments 44-60, wherein the anti-tau antibody comprises a VL domain that is encoded by a nucleic acid comprising at least 80%, at least 85%, at least 90%, at least 95% sequence identity to a sequence selected from SEQ ID NOs: 57-62. Numbered embodiment 62 comprises the anti-tau antibody of numbered embodiments 44-61, wherein the anti-tau antibody comprises a VH domain that is encoded by a nucleic acid comprising at least 80%, at least 85%, at least 90%, at least 95% sequence identity to a sequence selected from SEQ ID NOs: 52-56 and a VL domain that is encoded by a nucleic acid comprising at least 80%, at least 85%, at least 90%, at least 95% sequence identity to a sequence selected from SEQ ID NOs: 57-62. Numbered embodiment 63 comprises the anti-tau antibody of numbered embodiments 44-62, wherein the anti-tau antibody comprises a VH domain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NOs: 52-56. Numbered embodiment 64 comprises the anti-tau antibody of numbered embodiments 44-63, wherein the anti-tau antibody comprises a VL domain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NOs: 57-62. Numbered embodiment 65 comprises the anti-tau antibody of numbered embodiments 44-64 wherein the anti-tau antibody comprises a VH domain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NOs: 52-56 and a VL domain that is encoded by a nucleic acid comprising a sequence identical to SEQ ID NOs: 57-62.

EXAMPLES

[0134] The following examples are given for the purpose of illustrating various embodiments of the invention and are not meant to limit the present invention in any fashion. The present examples, along with the methods described herein are presently representative of preferred embodiments, are exemplary, and are not intended as limitations on the scope of the invention. Changes therein and other uses which are encompassed within the spirit of the invention as defined by the scope of the claims will occur to those skilled in the art.

Example 1: Tau Antibody Screening

[0135] Tau antibodies that detect phosphorylated Tau were assayed in the Simoa® bead assay using a 2-step or 3-step protocol assay according to manufacturer's instructions. See FIG. 1.

[0136] The antibodies tested were Antibody 1, Antibody 2, Antibody 3, Antibody 4, Antibody 5, and Antibody 6. The capture results can be seen in FIGS. 2A-2D. FIG. 2A is data from a 2-step protocol assay in which all capture antibodies were tested against Tau-12 detector (detects Tau at the N-terminus). The data demonstrates all captures had similar results with improved sensitivity seen with Antibody 6. FIG. 2B is data from a 2-step protocol assay in which all capture antibodies were tested against HT7-BT2 detectors (detects Tau in the mid-domain region). Both biotinylated antibodies were used. Data shows about a 10-fold increase in background compared to Tau-12 detector. No signal at 1000 µg/mL for any capture was detected. FIG. 2C is data from a 3-step protocol assay in which all capture antibodies were tested against Tau-12 detector. The data demonstrates reduced sensitivity compared to the 2-step protocol and improved sensitivity seen with Antibody 6. FIG. 2D is data from a 3-step protocol assay in which all capture antibodies were tested against HT7-BT2 detectors. Both biotinylated antibodies were used. Data shows about a 10-fold increase in background compared to Tau-12 detector. Based on the results, the 2-step protocol was then further optimized for sensitivity.

Example 2. Pharmacokinetics of Tau Antibodies

[0137] Antibodies were tested for pharmacokinetic profile.

[0138] The antigen information for the antibodies is seen in Table 10.

TABLE-US-00010	TABLE	10	Descrip-	Antigen	SEQ	tion	Name	ID	NO:	Sequence
WZN-1A	74	RSRTPSLP(PT)PPTREPKC	pT217b	WZN-1B	75	TPSLP(PT)PPTREPKKVAC	T217			

WZN-1C 76 RSRTSLPTTPREPKKQAC pT212 WZN-1D 77 RSR(pT)PSLPTPPTREPCKC
pS214 WZN-1E 78 RSRT(pS)LPTPPTREPCKC pT220 WZN-1F 79
RSRTPSLPTPP(PT)REPCKKQAC pT231 WZN-1G 80 KQAVVR(pT)PPKSPSSAC T231 WZN-1H
81 KQAVVRTPPKSPSSAC

[0139] Antibodies were generated and purified. The antibodies were assayed using a standard indirect ELISA protocol. Briefly, peptide antigens corresponding to SEQ ID NOs: 74-81 were diluted to 1 µg/ml in PBS and plated onto a Greiner Bio One Microton 96 well plate. Peptide antigens were produced by the manufacturer Abcam. WZN-1A and WZN-1B served as targets. WZN-1C, WZN-1D, WZN-1E, WZN-1F, WZN-1G, and WZN-1H served as negative controls. In the peptide antigen sequences, a phosphorylated residue is indicated by (pT) for phosphorylated threonine or (pS) for phosphorylated serine. After blocking with 1% BSA in PBS pH 7.4, antibodies were serially diluted 1 to 4 with an initial concentration of 1 µg/ml. After incubation, unbound antibodies were washed off with 1×TBST and HRP labeled goat anti-rabbit secondary antibody was applied according to the manufacturer's instructions. Subsequently, unbound secondary antibody was washed off with 1×TBST and 3,3',5,5'-tetramethylbenzidine (TMB) was applied for 5 minutes at room temperature and plates were read at 650 nm. Data is seen in FIG. 3. FIG. 3 shows screening data of the different monoclonal antibodies to varying peptide concentrations.

Example 3. Tau Antibodies for Immunohistochemistry

[0140] Tau antibodies described herein were tested in immunohistochemistry assays.

[0141] Briefly, all antibodies were optimised using a range of concentrations (0.01-3.00 µg/ml) and stained using a Leica Bond RX automated IHC platform: ER1 antigen retrieval (sodium citrate, pH 6) 20 mins at 100° C.; primary antibody 15 minutes at RT; IVD grade Leica Polymer Refine HRP detection 8 minutes at room temperature; DAB chromogen 10 minutes at room temperature, and finally hematoxylin counterstain 5 minutes at room temperature. Antibodies that passed basic IHC staining went on to undergo IHC staining following alkaline phosphatase (AP) treatment (200 U/ml, 37° C. for 60 minutes). A vehicle-only control (buffer containing no AP) was also employed. Positive antigen control tissues were FFPE normal human cerebral cortex and cerebral cortex from an Alzheimer patient. Negative antigen control tissues were FFPE normal human liver, skeletal muscle and heart muscle. All tissues were collated into a tissue micro array to streamline the IHC staining process. Negative reagent (detection system only) controls were employed and shown to be negative. Benchmark antibodies stained alongside the test antibodies were rabbit monoclonal [EPR22524-95] to Tau (ab254256, Abcam plc) and rabbit monoclonal [EPR1884(2)] to Tau (phospho S214) (ab170892, Abcam plc).

[0142] The benchmark antibodies demonstrated that the antibodies exhibited negative staining in the negative control tissue and positive staining in the positive control tissue (data not shown). Data for the Tau antibodies is seen in FIGS. 4A-4G (Antibody 6), FIGS. 5A-5G (Antibody 5), and FIGS. 6A-6G (Antibody 2). Antibody 5 and Antibody 6 demonstrated similar data as benchmark antibodies in that negative staining was observed in negative control tissues that included normal heart, normal liver, and normal skeletal muscle, and normal cerebral cortex and positive staining was observed in positive control tissues that included Alzheimer cerebral cortex. Antibodies 1-4 did not exhibit similar data as benchmark antibodies.

Example 4. Detection of Tau Peptides Using Tau Antibodies

[0143] Tau antibodies described herein that detect phosphorylated Tau were tested in ELISA assays. Tau antibodies were first tested for pTau 217 reactivity by indirect ELISA. FIG. 7 displays a diagram depicting the indirect ELISA assay format utilized. Briefly, streptavidin beads were bound to a plate and biotinylated peptide was added to the plates under conditions allowing for biotin-streptavidin binding. The biotinylated peptide was a synthetic peptide comprising a portion of Tau and possessing a phosphorylated threonine residue at position 217 (pT217). This was the target peptide. After binding of the synthetic peptide to the plate by the formation of the biotin-

streptavidin complex, Tau antibodies were added to the plates under conditions allowing for antibody-target peptide binding. After binding, plates were washed to remove any unbound antibody and plates then had a secondary antibody, or a tracer antibody, added (either goat anti-mouse antibody conjugated to peroxidase or donkey anti-rabbit antibody conjugated to peroxidase) directed to the species from which the Tau antibodies were derived. After binding, plates were washed to remove any unbound tracer antibody and plates next had TMB ELISA Peroxidase chromogenic substrate (3, 3', 5, 5'-Tetramethylbenzidine) added to visualize antibody reactivity in indirect ELISA experiments. Antibody sample binding was quantitated using an ELISA microplate reader.

[0144] As shown in FIG. 7, five antibodies were tested for an ability to detect phosphorylated Tau using this indirect ELISA technique. IBA493 mAb corresponds to a rabbit anti-Tau antibody capable of binding to Tau phosphorylated at threonine residue 217 (pTau 217) (Eli Lilly and Company). PT3 corresponds to a mouse anti-phospho (T212/T217) Tau selective antibody (Janssen Biotech Inc.). 30H10L2 corresponds to Antibody 2 described herein. 71H1L2 corresponds to Antibody 6 described herein. 62H10L7 corresponds to Antibody 5 described herein. All five antibodies were assayed for reactivity with pTau 217 in two separate ELISA instruments at the following concentrations: 10.sup.-2, 10.sup.-1, 10.sup.-0, 10.sup.1, 10.sup.2, 10.sup.3, and 10.sup.4 ng/mL per plate. As can be seen in the Bio-pt655 (phospho T217) and Bio-pt660 (phospho T217) graphs, both IBA493 mAb and PT3 demonstrated a robust, concentration-dependent level reactivity to pTau 217. Antibody 2 demonstrated a more modest concentration-dependent level reactivity to pTau 217 revealed at 104 ng/mL per plate. Antibody 5 and Antibody 6 did not demonstrate reactivity to pTau 217 in these assays. A graph in FIG. 7 showing the results of this ELISA assay using the five test antibodies on phosphatase-treated pTau demonstrated the specificity of antibodies IBA493 mAb, PT3, and Antibody 2 in detecting phosphorylated Tau.

[0145] Tau antibodies described herein that detect phosphorylated Tau were tested in Simoa®-based assays. FIGS. 8-24 display results Simoa® assays designed to sensitive tests for Tau reactivity to antibodies described herein. In some aspects, Simoa®-based assays can be approximately 1000× more sensitivity at detecting a given analyte when compared to detection of the same analyte in an indirect ELISA assay. This elevated sensitivity of Simoa®-based assays allows for the development and use of biomarkers that previously could not have generated a detectable signal using a traditional assay such as indirect ELISA. The elevated sensitivity of Simoa®-based assays when compared to conventional immunoassays such as indirect ELISA is due to the fact that the Simoa® method is capable of detecting single target molecules whereas conventional immunoassays typically require large reaction volume and millions of fluorophores, or millions of antibody-conjugated enzymes reacted to a color-producing substrate, before an optical signal can be detected. For Simoa®-based assays, average enzyme per bead (AEB) denotes raw signal output.

[0146] In FIG. 8, Antibody 2, as described herein and is capable of detecting pTau 217, was used in a Simoa®-based assay to detect a level of an analyte per plasma sample derived from an individual. Signal-to-noise (S/N) ratio was determined by Simoa® for each sample and plotted in a graph. 120 plasma samples were taken from individuals and assayed. The graphed S/N ratio indicated that all tested samples apart from one yielded a signal within the expected concentration range. When plasma samples were diluted 1:3 and then assayed again, only 3 of the 120 samples yielded a result below measurement of a blank control and only 5 samples registered a measurement of S/N 1.5, which was determined to be the limit of detection (LOD). In FIG. 8, with each of the 120 plasma samples assayed, a calculation was made to determine the coefficient of variation (CV %) for each sample and the results were graphed against measured concentration. 10 of the 120 samples yielded a CV % greater than 20 and from this analysis, the estimated analytical lower limit of quantitation (LLOQ) was determined to be 0.08 pg/mL. This LLOQ value represents the lowest amount of an analyte (Tau phosphorylated at T217) that can be quantitatively determined with an acceptable

level of precision. These results in FIG. 8 indicated the sensitivity of the Simoa® method to detect Tau phosphorylated at T217 using Antibody 2.

[0147] In FIG. 9, calibration curves were generated for the Simoa® pTau-217 assay and graphed [AEB vs log(CAL) pg/mL] using 68 CSF samples and 120 plasma samples separated into 4 plates using Antibody 2 and 4 plates using ADx Neuroscience antibody ADx204. In another graph from this assay performed on a separate instrument [AEB vs log(CAL) pg/mL] with AEB plotted on a log scale demonstrated the fit of the data can enable accurate analyte quantitation calculations when measuring samples.

[0148] In FIG. 10, 38 paired CSF and EDTA plasma samples were measured with the Simoa® pTau-217 assay using Antibody 2. This assay is also named as ALZpath Dx. Results were graphed and samples were indicated with their clinical diagnosis (non-AD, uncertain, or AD). These results, and the statistical analysis thereof, indicated a strong correlation between CSF and plasma pTau levels as measured with the Simoa® pTau-217 assay using Antibody 2 (R value ~0.7 and P value for two-tailed T test <0.0001 between non-AD and AD).

[0149] In FIG. 11, 42 sCSF samples were measured with the Simoa® pTau-217 assay using Antibody 2 and a Simoa® pTau-181 assay using a pTau-181 antibody from Quanterix® (Quanterix® Corp., Item number 103714) and plotted against each other. This demonstrated that the Simoa® pTau-217 assay using Antibody 2 showed the expected relationship with an analyte implicated in AD detected in CSF (pTau-181). Statistical analysis indicated an R value ~0.8 and P value for two-tailed T test <0.0001.

[0150] In FIG. 12, 42 CSF samples were measured with the Simoa® pTau-217 assay using Antibody 2 and a Simoa® pTau assay using Innostest pTau-181 antibody and plotted against each other. This demonstrated that the Simoa® pTau-217 assay using Antibody 2 showed the expected relationship with an analyte implicated in AD detected in CSF (pTau). Statistical analysis indicated an R value ~0.77 and P value for two-tailed T test <0.0001.

[0151] In FIG. 13, 42 CSF samples were measured with a Simoa® HD-X assay using Antibody 2 as a capture antibody, ADx204 antibody as a detector, and a peptide as calibrator. This demonstrated that the Simoa® assays using known AD biomarkers showed the expected relationship. Statistical analysis indicated an R value ~0.9 and P value for two-tailed T test <0.0001.

[0152] In FIG. 14, CSF samples and plasma samples were measured with the Simoa® pTau-217 assay using Antibody 2 and graphed in separate graphs. Clinical diagnosis of AD, or control with no AD diagnosis, was used as the classifier for each sample. Analysis of the graphed results indicated a significant difference between samples derived from individuals with a clinical AD diagnosis vs controls for both CSF samples and plasma samples. Area under the curve (AUC) calculation was 0.94 for CSF samples and 0.86 for plasma samples. These results indicated that the Simoa® pTau-217 assay using Antibody 2 was able to differentiate AD cases in CSF and plasma.

[0153] In FIG. 15, 4 EDTA plasma samples with high pTau levels serve as quality controls (labelled QC_L1, QC_L2, QC_M, and QC_H) were measured with the Simoa® pTau-217 assay using Antibody 2 in duplicate test and pTau levels were calculated. Plotting the results from repeated testing demonstrated the precision and reproducibility of the Simoa® pTau-217 assay using Antibody 2.

[0154] In FIG. 16, control samples from FIG. 15 and additional measured samples were measured with the Simoa® pTau-217 assay using Antibody 2 and plotted in two separate experiments to generate precision profiles. The precision profiles are based on measured sample concentration and inter-run CV % of the 4 QC samples. From this experiment, a functional LLOQ of pTau-217 in this assay was determined to be 0.26 pg/mL.

[0155] In FIG. 17, parallelism was assessed in the Simoa® pTau-217 assay using Antibody 2. A determination of parallelism is also important in that it shows if a signal is specific. Parallelism determines whether actual samples containing high endogenous analyte concentrations provide the same degree of detection in the assay in a standard curve after dilutions. This can represent

differences in antibody binding affinity to endogenous analyte and a standard or calibration analyte. This can ensure that recombinant standards parallel native recognition of the endogenous analyte. In this experiment, 4 plasma samples, each from different donors, with relatively high concentration of detected pTau-217 and a spiked dilution buffer (sample 5) were diluted with a factor of 2 in 5 steps, starting at a dilution of 3×. The concentrations dropped below LLOD from dilution factor 12× onwards for all 4 plasma samples. In a graph of log (measured pg/mL) vs log [dilution factor (DF)] plasma measurements over spike measurements demonstrated linearity in detection along the various dilutions. 3 out of the 4 plasma samples were determined to fall within the accepted range of parallelism, with Sample 4 falling just outside of the accepted range. The accepted range of parallelism is <15%. These results demonstrated that the Simoa® pTau-217 assay using Antibody 2 on plasma samples yielded consistent and precise calculations of pTau-217 levels across various concentrations thus demonstrated its utility as a biomarker assay.

[0156] In FIGS. **18-19**, dilution linearity using the Simoa® pTau-217 assay using Antibody 2 was performed to demonstrate that a sample with a spike concentration about the upper limit of quantification (ULOQ) can be diluted to a concentration within the working range while still yielding a reliable assay result. In FIG. **18**, three spiked samples (s1, s2, and s3) and the Calibration sample were assayed and plotted as log(measured pg/mL) vs log (DF) to determine dilution linearity. In FIG. **19**, three spiked samples (s1, s2, and s3) and the Calibration sample were assayed and plotted as log(measured pg/mL) vs log (DF) to determine dilution linearity with the highest spike point omitted from s1, s2, and s3 since it was out of the calibration range of 50 pg/mL.

[0157] In FIG. **20**, the Simoa® pTau-217 assay using Antibody 2 was used to assess samples taken from a memory clinic cohort. Plasma samples were measured and graphed for calculated pTau217 concentrations. Clinical diagnosis of AD was used as the classifier. AUC was calculated at 0.916 indicating success in distinguishing AD+ from AD- within this cohort with the Simoa® pTau-217 assay using Antibody 2. Also in FIG. **20**, a receiver operating characteristic (ROC) curve was plotted to illustrate the diagnostic ability of this binary classifier (AD+ or AD-) system as it is possible that the discrimination threshold between classifiers is varied.

[0158] In FIG. **21-22**, clinical performance of the Simoa® pTau-217 assay using Antibody 2 was compared to a Simoa® pTau-181 assays using antibody P-tau181-Quanterix®. In FIG. **21**, Antibody 2 was able to distinguish assayed plasma samples from taken from AD dementia individuals vs Controls (P value 1.3e.sup.-12 for Antibody 2). In FIG. **22**, the commercially available P-tau181-Quanterix® Simoa® assay (Quanterix® Corp., Item number 103714) was also able to distinguish assayed plasma samples from taken from AD dementia individuals vs Controls (P value 9.6e.sup.-08). Data from the individuals from which the samples were derived for data from FIGS. **21-22** is listed in FIG. **22**.

[0159] In FIG. **23** precision plots were generated for the Simoa® assay using P-tau217 Antibody 2 and P-tau181-Quanterix® antibody (Quanterix® Corp., Item number 103714). Calculated LLODs were 0.55 pg/mL and 0.24 pg/mL, respectively. Concentrations were not back calculated and LLOD values are accordingly not back calculated.

[0160] In FIG. **24** ROC curves were plotted for the P-tau181-Quanterix® antibody (Quanterix® Corp., Item number 103714) and P-tau217 Antibody 2. Analysis of the data indicated that P-tau217 Antibody 2 exhibited superior sensitivity and specificity when comparing to the Simoa® assay using the P-tau181-Quanterix® antibody in differentiating AD-dementia from controls. The diagnostic accuracy of Antibody 2 for AD dementia in this Simoa® method is 92.5% when tested on plasma samples. The diagnostic specificity of Antibody 2 for AD dementia in this Simoa® method is 85% when tested on plasma samples.

[0161] In FIG. **25** a schematic diagram of Tau polypeptide indicating the relative location of various protein domains and the locations of threonine residues which can be assayed for phosphorylation status using methods disclosed herein is depicted. The location of pT217 within the P2 domain of Tau is indicated. pT181 resides within the P1 domain and pT231 resides near the

border between the P2 and RI domains.

[0162] In FIG. **26** various Tau antibodies were assayed using indirect ELISA and extent of reactivity to a Tau fragment with non-phosphorylated T217 (Bio-pt654) and full length Tau (Tau441) were graphed. IBA493 mAB and PT3 displayed concentration dependent reactivity to both Bio-pt654 and Tau441. Antibody 2, 5, and 6 described herein do not display any reactivity to either Bio-pt654 or Tau441 in this assaying demonstrating precision and specificity in pTau-217 detection for Antibodies 2, 5, and 6.

[0163] In FIG. **27** various Tau antibodies were assayed using indirect ELISA and extent of reactivity to a Tau fragment with phosphorylated T181 (Bio-pt126) and phosphorylated T231 (Bio-pt146) were graphed. IBA493 mAB and Antibody 2 described herein displayed concentration dependent reactivity to Bio-pt126. IBA493 mAB was the only antibody of those tested displaying concentration dependent reactivity to Bio-pt146. This demonstrates that IBA493 mAB, PT3, and Antibody 2 described herein were each distinguishable based on which analytes each antibody interacted with via indirect ELISA. IBA493 mAB interacts with pTau-217, non-phospho T217, full length Tau, pTau-181, and pTau-231. PT3 interacts with pTau-217, non-phospho T217, and full length Tau. Antibody 2 interacts with pTau-217 and pTau-181. IBA493 mAB's interaction with non-phospho T217 was also shown to be considerably less than PT3's interaction with non-phospho T217.

[0164] In FIG. **28**, a diagram of an assay using Antibody 2 to detect capture of particular Tau peptides is depicted. In this assay, Antibody 2 is bound to a plate and sample wells from the plate are subjected to various biotinylated peptides under conditions conducive to forming specific antibody-ligand interactions. Samples are then washed to remove excess unbound biotinylated peptide. Streptavidin beads conjugated to peroxidase and then added to the samples to allow biotin-streptavidin complexes to form on peptide bound antibodies. TMB.sub.red substrate is then added and samples are measured for colorimetric development using an ELISA plate reader. The results were graphed and out of various pTau-217, pTau-231, and pTau-181 peptides tested, Adx-pt655 yielded specific dose-dependent reactivity. These results illustrate the specificity of Antibody 2 to specific features of pTau (namely Tau phosphorylated at threonine 217). Antibody 5 and Antibody 6 tested by indirect ELISA under the same conditions using the same Tau peptides yielded no specific dose-dependent reactivity toward the Tau peptides tested.

[0165] Various pTau-217 antibodies corresponding to Antibody 1, Antibody 2, Antibody 3, Antibody 4, Antibody 5, and Antibody 6 described here were also evaluated as capture antibodies either directly coated onto or onto streptavidin-coated plates on the Mesoscale Discovery technology platform. This system uses non-radioactive, electrochemiluminescent labels, thereby conferring significant advantages over traditional ELISA assays. These advantages include lower background signal, improved sensitivity, and a dynamic range of detection.

[0166] In FIG. **29**, Western blots were used to assess binding of various antibodies to brain lysate samples from AD patients and Control subjects. In the five Western blots shown, samples are loaded according to the same sample key shown in FIG. **29**. Protein ladders were run in lanes 1 and 10. Phosphatase treated pTau loaded in an amount of 0.05 ug was run in lane 2. Full-length Tau (Tau411) loaded in an amount of 0.05 ug was run in lane 3. Lanes 4-6 contain samples from different Control subjects with a dilution factor of 5. Lanes 7-9 contain samples from different AD subjects with a dilution factor of 5. The results indicated that IBA394 mAb and PT3 both bound to and immunoprecipitated different length isoforms of Tau in control samples and AD samples and immunoprecipitated significantly more Tau in AD samples while showing no interaction with synthetic full length Tau or phosphatase treated pTau. Antibody 2 (30H2L10) bound to and immunoprecipitated different length isoforms of Tau in AD samples but did not immunoprecipitate a significant amount of Tau in samples from Control individuals. Antibody 5 and Antibody 6 did not yield detectable Western blot signals in this assay.

[0167] While preferred embodiments of the present disclosure have been shown and described

herein, it will be obvious to those skilled in the art that such embodiments are provided by way of example only. Numerous variations, changes, and substitutions will now occur to those skilled in the art without departing from the disclosure. It should be understood that various alternatives to the embodiments of the disclosure described herein may be employed in practicing the disclosure. It is intended that the following claims define the scope of the disclosure and that methods and structures within the scope of these claims and their equivalents be covered thereby.

Claims

1. A nucleic acid encoding an antibody or antigen-binding fragment that binds to a phosphorylated tau polypeptide, comprising: a. a nucleic acid sequence encoding a heavy chain of the antibody or antigen-binding fragment, wherein the heavy chain comprises a variable heavy chain (VH) domain comprising HCDR1 sequence comprising SEQ ID NO: 2, HCDR2 sequence comprising SEQ ID NO: 7, and HCDR3 sequence comprising SEQ ID NO: 11; or b. a nucleic acid sequence encoding a light chain of the antibody or antigen-binding fragment, wherein the light chain comprises a variable light chain (VL) domain comprising LCDR1 sequence comprising SEQ ID NO: 15, LCDR2 sequence comprising SEQ ID NO: 21, and LCDR3 sequence comprising SEQ ID NO: 25.
2. The nucleic acid of claim 1, wherein the nucleic acid sequence encoding the heavy chain of the antibody or antigen-binding fragment encodes a VH domain comprising SEQ ID NO: 31.
3. The nucleic acid of claim 1, wherein the nucleic acid sequence encoding the light chain of the antibody or antigen-binding fragment encodes a VL domain comprising SEQ ID NO: 36.
4. The nucleic acid of claim 1, wherein the nucleic acid sequence encoding the heavy chain of the antibody or antigen-binding fragment encodes a heavy chain comprising SEQ ID NO: 43.
5. The nucleic acid of claim 1, wherein the nucleic acid sequence encoding the light chain of the antibody or antigen-binding fragment encodes a light chain comprising SEQ ID NO: 44.
6. The nucleic acid of claim 1, wherein the nucleic acid sequence encoding the heavy chain of the antibody or antigen-binding fragment is at least 80% identical to SEQ ID NO: 53.
7. The nucleic acid of claim 1, wherein the nucleic acid sequence encoding the light chain of the antibody or antigen-binding fragment is at least 80% identical to SEQ ID NO: 58.
8. The nucleic acid of claim 1, wherein the nucleic acid sequence encoding the heavy chain of the antibody or antigen-binding fragment is at least 80% identical to SEQ ID NO: 64.
9. The nucleic acid of claim 1, wherein the nucleic acid sequence encoding the light chain of the antibody or antigen-binding fragment is at least 80% identical to SEQ ID NO: 69.
10. A vector comprising the nucleic acid of claim 1.
11. The vector of claim 10, wherein the vector is an expression vector.
12. The vector of claim 10, wherein the vector is a recombinant vector.
13. A host cell comprising the nucleic acid of claim 1.
14. A host cell comprising the vector of claim 10.
15. A method of making the antibody or antigen-binding fragment that binds to the phosphorylated tau polypeptide, the method comprising: (1) introducing the nucleic acid of claim 1 directly into a host cell, (2) culturing the host cell, and (3) isolating the antibody or antigen-binding fragment.
16. The nucleic acid of claim 1, wherein the phosphorylated tau polypeptide is selected from the group consisting of pTau-181, pTau-212, pTau-217, pTau-231, pTau-214, and pTau-220.
17. The nucleic acid of claim 1, wherein the phosphorylated tau polypeptide is pTau-181, pTau-217, or pTau-231.
18. The nucleic acid of claim 1, wherein the antibody or antigen-binding fragment comprises an IgG-scFv, nanobody, BiTE, diabody, DART, TandAb, scDiabody, scDiabody-CH3, triple body, mini-antibody, minibody, TriBi minibody, scFv-CH3 KIH, Fab-scFv-Fc KIH, Fab-scFv, scFv-CH-CL-scFv, Fab', F(ab')₂, F(ab')₃, F(ab')₂-scFv₂, scFv, scFv-KIH, Fab-scFv-Fc, tetravalent HCAb, scDiabody-Fe, diabody-Fe, tandem scFv-Fc, or intrabody.

19. The nucleic acid of claim 1, wherein the antibody or antigen-binding fragment is a monoclonal antibody.

20. The nucleic acid of claim 1, wherein the antibody or antigen-binding fragment is a rabbit monoclonal antibody.
